

Blameocracy: Causal Rhetoric in Politics*

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Abstract

This paper studies causal rhetoric in politics: language that links political actions to outcomes through blame and credit. We measure its use in a corpus of 4.2 million congressional tweets spanning 2012 to 2023 and analyze its economics. We document that causal rhetoric becomes increasingly central in politicians' communication. Leveraging quasi-exogenous variation in Twitter penetration, we show that causal rhetoric—especially blame—generates political returns in small-donor contributions: blame alone mobilizes additional donors, whereas credit increases only average giving. Using fine-grained data on legislative activity, we find that politicians strategically reallocate blame and credit as policy ownership shifts their relative credibility. Finally, economic causal rhetoric appears only weakly grounded in mainstream macroeconomics, especially blame.

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1 Introduction

Democratic politics hinges on how citizens link political actions to outcomes: who is responsible for what happened, and what consequences political actors are expected to produce in the future (Schmitter and Karl, 1991). Yet the relationship between political actions and collective outcomes is rarely unambiguous. Policies operate in complex environments, outcomes have many causes, and political responsibility is often diffuse. As a result, citizens may struggle to reconstruct the mapping from political actions to outcomes, making it difficult to infer both the causes of what they observe and the future consequences of those actions (Achen and Bartels, 2017).

This difficulty creates scope for politicians to shape voters’ accounts of these connections through causal rhetoric—language that maps political actions to outcomes through blame and credit. Crucially, this rhetorical margin may be especially consequential because causal framing has been shown to be highly persuasive, shaping what information people seek, remember, and act upon (Andre et al., 2024; Bursztyrn et al., 2023; Graeber et al., 2024; Hüning et al., 2022).¹

Yet existing work has mainly examined how accountability incentivizes politicians, how voters evaluate incumbents, or whether campaigns attack opponents or promote the self, while largely overlooking politicians’ use of causal language linking political actions to outcomes (Ashworth, 2012; Ferraz and Finan, 2025; Healy and Malhotra, 2013; Lau and Rovner, 2009). As a result, we still know little about politicians’ use of causal rhetoric, the returns it generates, and how politicians strategically deploy it.

This paper studies causal rhetoric in politics. Using natural language processing, we measure it in a corpus of 4.2 million tweets posted by U.S. Members of Congress between 2012 and 2023. First, we document how causal rhetoric becomes increasingly central over our sample period, rising pervasively. Then, leveraging quasi-random variation in early Twitter adoption, we show that causal rhetoric—especially blame—has sizable effects on campaign contributions from small donors: blame alone mobilizes additional donors, whereas credit increases only the average amount given per donor. Using fine-grained data on legislative activity, we also find that politicians use blame and credit strategically, reallocating between the two as policy ownership shifts their relative credibility: politicians blame more when the opposing party acts and

¹More broadly, a rich literature in cognitive psychology emphasizes that human reasoning is fundamentally organized around causal thinking (e.g., Chater and Loewenstein, 2016; Lombrozo and Vasiljeva, 2017; Sloman and Lagnado, 2015).

claim more credit when their own party does. Finally, using a structured AI-assisted coding procedure, we find that economic causal rhetoric, and blame in particular, often appears only weakly grounded in mainstream macroeconomics.

A central challenge in our analysis is measurement. In natural language, links between political actions and outcomes are conveyed through causal attributions, which are often implicit rather than systematically signaled by standard lexical or syntactic markers, especially in political discourse. This makes them difficult to detect with standard dictionary- or syntax-based tools. We address this limitation using a supervised learning approach built on a bidimensional classification of tweets. A tweet is coded as causal if it links the action of a political actor to an outcome. Separately, we classify its tone as positive, negative, or neutral, based on the attitude expressed toward its subject. Combining these two dimensions yields our measure of rhetorical style: causal tweets with negative tone are labeled as blame, those with positive tone as credit, and all remaining tweets as none.

To implement this measure at scale, we hand-label roughly 4,000 tweets. As part of this process, we jointly annotate about 500 tweets to assess reliability, obtaining substantial agreement, with a Fleiss' Kappa of 0.64. We then fine-tune a RoBERTa-large model, pre-trained on 154 million tweets, to classify tweets as blame, credit, or none. The resulting classifier achieves strong predictive performance, with 0.83 accuracy, 0.84 F1-score, and 0.73 Matthews Correlation Coefficient.

We validate the measure through both external and internal checks. Externally, we compare our labels with independent tweet-level classifications of policy-attack and credit-claiming from America's Political Pulse ([Westwood and Lelkes, 2024](#)). In a matched sample of 101,689 tweets, our blame label agrees with the corresponding policy-attack label in 87 percent of cases, with a correlation of 0.62, while our credit label agrees with the corresponding credit-claiming label in 84 percent of cases, with a correlation of 0.60. The alignment is similarly strong at the politician level: across the 443 politicians observed in both datasets over the common sample period, the share of blame tweets correlates 0.85 with the share of policy-attack tweets, and the share of credit tweets correlates 0.78 with the share of credit-claiming tweets. Internally, blame tweets in our data are more outward-directed and more retrospective, whereas credit tweets are more self-directed and more forward-looking. Finally, we show that blame and credit are not reducible to existing measures of sentiment, emotionality, or moral language, indicating that they capture a distinct rhetorical dimension.

Our measurement framework allows us to document three key facts about the supply of causal rhetoric in politicians' communication on social media. First, causal rhetoric becomes increasingly central over our sample period. Blame and credit together account for about 19 percent of tweets in 2012, with both blame and credit

at about 10 percent, but rise to roughly 43 percent by 2023, with both converging at around 20 percent.

Second, the rise is pervasive. An event-study design shows that the increase persists after controlling for both politician and topic fixed effects: accounting for composition absorbs less than one fourth of the rise in blame and amplifies the rise in credit by about one fifth. The increase is broadly distributed across politicians, but disproportionately concentrated in policy domains.

Third, as causal rhetoric becomes more central, it increasingly crowds out purely affective messaging—that is, positive or negative tweets that do not contain causal claims. The share of blame among negative tweets rises from 42 to 58 percent, while the share of credit among positive tweets rises from 14 to 38 percent. At the same time, causal rhetoric becomes the main vehicle through which elite polarization is expressed. By the Biden presidency, partisan differences in sentiment are close to zero in non-causal tweets, but approach a full standard deviation in tweets classified as blame or credit.

To study causal rhetoric in economic terms, we interpret its use through the lens of a production problem (Aridor et al., 2024), in which politicians choose rhetorical inputs based on both the returns they generate and their relative convenience in a given political context. A natural return of interest is political support. Yet support is difficult to observe at the frequency at which politicians communicate on social media. We therefore turn to campaign contributions from small donors, which provide a high-frequency revealed-preference measure of political support and political mobilization (Bouton et al., 2022; Petrova et al., 2021). Beyond total revenue, donation data also allow us to distinguish two margins of support: the number of donors, which captures mobilization, and the average amount per donor, which captures support deepening.

To identify the causal effect of blame and credit on donations, we exploit cross-county variation in Twitter penetration across the United States. In counties with greater Twitter penetration, residents are more likely to be exposed to politicians' rhetorical content. We therefore estimate whether monthly donations from a given county to a given politician respond differentially to that politician's monthly use of blame and credit as a function of local Twitter penetration. To address the endogeneity of local Twitter adoption, we instrument county-level Twitter penetration with the early diffusion of Twitter following the 2007 South-by-Southwest festival, using the number of followers of the official SXSW Twitter account who joined the platform in 2007 (Müller and Schwarz, 2023).

We find that both blame and credit increase support from small donors, but they do so in different ways. At the aggregate level, blame has a much larger effect: a one-standard-deviation increase in the monthly share of blame tweets raises donation

revenue by about 3.2 percent in a county with average Twitter penetration, whereas the corresponding effect of credit is only about 0.5 percent.

Decomposing these effects reveals distinct margins of support. Blame is the only rhetorical style that expands support at the extensive margin, by increasing the number of donors, whereas credit has essentially no effect on this margin. In particular, a one-standard-deviation increase in the monthly share of blame tweets posted raises the number of donors by about 1.6 percent in a county with average Twitter penetration. Credit, in turn, operates only at the intensive margin, by increasing the average amount per donor: a one-standard-deviation increase in the monthly share of credit tweets posted raises the average amount given by about 0.5 percent in a county with average Twitter penetration. Blame also increases average giving, indicating that its effects are not confined to the extensive margin. Taken together, these patterns suggest that blame alone mobilizes support, whereas credit only deepens it.

Consistent with this interpretation, we also show that blame spreads more virally: blame tweets receive substantially more retweets than non-causal tweets, are disproportionately likely to reach the upper tail of the retweet distribution, and, while accounting for only about 15 percent of tweets, generate nearly 40 percent of all retweets.

These effects also differ across ideological constituencies. Classifying donors within each electoral cycle as moderate or extreme depending on whether the absolute value of their ideological score falls below or above the cycle-specific median, we find that extreme donors respond positively to blame and negatively to credit, whereas moderate donors display the opposite pattern. The symmetry is only qualitative, however: the positive response of extreme donors to blame is substantially larger than the positive response of moderate donors to credit. Together, these findings show that, despite blame being more effective than credit in the aggregate, the two appear politically complementary: they operate through different margins and appeal to different segments of the donor base.

To study the strategic use of blame and credit, we next turn to choice data. If politicians deploy causal rhetoric strategically, their use of blame and credit should reflect not only the returns these rhetorics generate on average, but also the political conditions under which each can be used most convincingly. Raw choice data already show that politicians do not all rely on the same rhetorical mix, and that they do not simply lean exclusively on blame despite its stronger average returns, suggesting a non-trivial allocation problem. More informative still, this allocation varies systematically with institutional position: politicians in opposition rely relatively more on blame, whereas politicians in power rely relatively more on credit. This pattern suggests that credibility is a key force shaping strategic use. Because causal claims must remain tied to observable political reality to persuade, policy ownership changes which of the two

rhetorics is more credible: blaming is more credible when the opposing party is responsible for political action, whereas claiming credit is more credible when one's own party is.

To assess this mechanism more directly, we use bill introductions in Congress as a fine-grained proxy for policy ownership. The pattern is sharp. Blame tweets are about 5 percentage points more likely to be posted when the opposing party introduces a bill, whereas credit tweets are about 6 percentage points more likely when the politician's own party does. These results show that politicians strategically reallocate blame and credit as policy ownership changes, whichever of the two claims is more credible.

Finally, we assess the content of the causal claims embedded in politicians' rhetoric. The political returns documented above need not reflect voters linking political actions to outcomes more accurately. They may instead arise because politicians advance causal claims that are weakly grounded or that omit qualifications in ways that tilt the attribution of outcomes in their favor. We focus on blame and credit tweets about the economy, where such claims can be benchmarked against a relatively well-defined body of mainstream macroeconomic knowledge. Using a structured AI-assisted coding procedure, we classify the dominant causal claim in each economic tweet according to how strongly it is supported by standard macroeconomic reasoning. We find that most claims are not cleanly grounded: about 90 percent require either minor or major qualifications, while fewer than 10 percent are supported as presented. Moreover, blame is systematically less grounded than credit, despite being the most politically effective rhetorical style. These findings suggest that causal rhetoric often convinces not by clarifying the link between political actions and outcomes, but by compressing complex economic causality into politically convenient attributions.

The rest of the paper proceeds as follows. We conclude the introduction by discussing the paper's contribution to the literature. Section 2 describes the data, and Section 3 presents our measure of blame and credit together with its validation. Section 4 documents the supply of causal rhetoric. Section 5 estimates its political returns, and Section 6 studies how politicians strategically allocate blame and credit. Finally, Section 8 concludes.

Related Literature

This paper speaks first to a long-standing literature on accountability in political economy. A large body of work studied how accountability shapes politicians' behavior, showing, for example, how term limits or press coverage discipline politicians' actions (Ashworth, 2012; Besley, 2006; Besley and Case, 1995; Ferraz and Finan, 2011; Snyder Jr and Strömberg, 2010). Another strand instead focused on how voters use re-

alized outcomes to evaluate incumbents in a retrospective-voting fashion ([Achen and Bartels, 2017](#); [Ferraz and Finan, 2008](#); [Healy and Malhotra, 2013](#)). We contribute by studying an earlier margin in the accountability process: how politicians may actively shape the attributions on which accountability rests. In particular, we document the use of rhetoric that links political actions to outcomes, and we study both the returns it generates for politicians and how they strategically allocate blame and credit across political contexts.

This focus on politicians' claims about how political actions and outcomes are connected also ties this paper to the literature on narratives. We add to a small set of papers that seek to measure narratives in observational data. Existing work has mainly focused on narratives about specific topics, such as climate change or stock-market crashes ([Gehring and Grigoletto, 2023](#); [Goetzmann et al., 2022](#); [Macaulay and Song, 2023](#)). We contribute by proposing a more agnostic and explicitly reduced-form approach: we identify whether politicians' language links political actions to outcomes, regardless of the topic under discussion. In this sense, our measure captures a broad rhetorical feature that traces reduced-form narratives mapping political actions into outcomes, and is therefore related to [Bilotta and Manferdini \(2026\)](#). In that paper, voters cannot fully recover the relationship between policies to outcomes, and politicians compete by offering narratives that map policies into outcomes by shifting blame and claiming credit. Our empirical evidence complements that framework by documenting the prevalence and returns of such rhetoric. More broadly, it also highlights the role of credibility in shaping its strategic use, an idea also shared by other theoretical work on narratives in political economy, where causal claims must remain at least minimally consistent with evidence ([Eliaz et al., 2025](#); [Eliaz and Spiegler, 2020](#)).

This paper also contributes to the literature on political rhetoric. A long-standing stream of work on negative campaigning studied whether campaigns attack opponents or promote the self ([Lau and Rovner, 2009](#)). We differ markedly by focusing on the causal margin of rhetoric: whether political language links political actions to outcomes. This leads us to study blame—negative rhetoric that maps political actors or their actions to bad outcomes—while abstracting from other negative content such as personal attacks, insults, or moral condemnation that contains no causal attribution. A more recent strand instead studied other expressive dimensions of political language, such as emotionality, emotions, and moral values ([Algan et al., 2025](#); [Enke, 2020](#); [Gennaro and Ash, 2022](#)). We contribute by isolating a neglected content dimension of political language: whether it connect political actions to outcomes.

Finally, this paper contributes to the literature on social media in political economy. A growing body of work shows that social media affects political support, political behavior, and political attitudes, and can also shape broader societal outcomes such as

protest activity (Enikolopov et al., 2020; Fujiwara et al., 2024; Gylfason, 2023; Müller and Schwarz, 2023; Rotesi, 2019; Zhuravskaya et al., 2020). Closest to our paper is a recent strand studying the effects of Twitter on campaign finance. Petrova et al. (2021) show that politicians benefit from adopting Twitter, especially entrants and candidates in high-penetration states, while Boken et al. (2023) document a fundraising premium when politicians go viral on Twitter. Relative to this work, we shift attention from platform adoption, diffusion, and virality to the persuasion strategies embedded in politicians’ messages. More broadly, we contribute by opening part of this black box: rather than treating social media exposure as a reduced-form treatment, we identify blame and credit as specific rhetorical inputs through which politicians’ online communication affects political support.

2 Data

Our analysis combines several datasets on political communication, campaign finance, and legislative activity. We merge these data with demographic and ideological information on elected officials.

Twitter Data Our main dataset contains approximately 4.2 million tweets posted by Democratic and Republican members of the U.S. Congress between 2012 and 2023. We combine data from the [CongressTweets](#) project and [Bellodi et al. \(2023\)](#). To the best of our knowledge, the resulting dataset covers all tweets posted by Representatives from January 3, 2013, to July 11, 2023; all tweets by Representatives running for re-election in 2012, over the full calendar year; and all tweets by Senators from June 21, 2017, to July 11, 2023. The final sample includes 4,198,455 tweets from 900 unique politicians and 1,789 unique Twitter accounts.² We enrich these data with demographic and ideological information from ProPublica, VoteView, and Wikipedia. Table [A1](#) reports descriptive statistics at both the tweet and politician levels. Republicans account for a slight majority of politicians in the sample, but most tweets are posted by Democrats, who—as also documented by [Fujiwara et al. \(2024\)](#)—are more active on Twitter. Most politicians in the dataset hold at least a bachelors degree, the average age is just under 60, and about one quarter are women.

Donations We collect campaign contribution data from the Database on Ideology, Money in Politics, and Elections compiled by [Bonica \(2024\)](#). This dataset includes more than 850 million donations made by individuals and organizations to candidates

²We exclude retweets because they do not represent original content. We retain quoted tweets because they also contain user-generated text.

in local, state, and federal elections between 1979 and 2024. We restrict the sample to individual contributions directed to candidates for the U.S. House of Representatives and U.S. Senate in the 2012–2024 election cycles. Following the literature, we define small donations as contributions below \$1,000 (Boken et al., 2023; Petrova et al., 2021). The resulting sample contains approximately 211 million unique donations. In the analysis, we link these donations to politicians in our Twitter data, so the effective sample period is determined by the overlap between donation records and observed tweeting behavior and therefore runs through 2023.

Bills We also construct a dataset of congressional bills using official records from the U.S. Government Publishing Office, accessed through the GovInfo bulk data repository. We scrape structured metadata for all bills introduced in the U.S. House and Senate from the 112th Congress (2011–2013) through the 118th Congress (2023–2025).³ This yields approximately 100,000 distinct legislative items. To focus on politically salient legislation, we restrict attention to bills sponsored by Democratic or Republican legislators that received at least one roll-call vote. The resulting sample contains 3,182 bills sponsored by 754 unique legislators between 2012 and 2024. In the analysis, we link these bills to politicians in our Twitter data, so the effective sample period is again determined by the overlap between legislative activity and observed tweeting behavior and therefore runs through 2023.

America’s Political Pulse We use data from America’s Political Pulse, a project led by the Polarization Research Lab that also tracks elite political rhetoric (Westwood and Lelkes, 2024). The project classifies congressional communications—including speeches, newsletters, tweets, press releases, and public statements—into rhetorical categories. These measures are available at the individual-politician level starting in 2023. We use all the tweets posted through the end of our sample period, yielding 158,745 tweets from 449 legislators.

3 Measuring Blame and Credit

A central challenge in our analysis is measurement. We need to determine, first, whether a tweet links the actual or potential action of a political actor to an outcome and, second, whether this takes a negative form (blame), a positive form (credit), or neither. Building on this two-dimensional definition, we construct a measure of blame

³Specifically, we include House bills, House joint resolutions, House concurrent resolutions, and House simple resolutions, as well as their Senate counterparts.

and credit at scale by fine-tuning a RoBERTa-large model. We then validate the resulting measure through a range of exercises, showing that the classifier performs well, that the measure aligns with both external benchmarks and internal linguistic patterns, and that the blame–credit dimension captures a distinct form of political rhetoric.

3.1 Definition

Our classification rests on two dimensions: *causality* and *tone*. A tweet is *causal* if it links the actual or potential action of a political actor to an outcome, and *tone* captures whether the tweet expresses a negative, neutral, or positive stance toward its subject.

Causality is binary. Political actors include politicians, institutions, and politically aligned organizations, but exclude natural events (e.g., pandemics or disasters) and neutral actors (e.g., scientific teams). A tweet is causal only if it links a politically meaningful action of such an actor to an outcome. The action may be explicit, as when the tweet refers to a particular bill, reform, or decision, or implicit, as when it attributes the outcome to the actor while leaving the specific action unstated. In either case, the action may be already taken, currently taken, or one that would be taken if the actor were in power. Outcomes may concern any economic consequence (e.g., income or inflation) or social consequence (e.g., civil liberties).

Tone captures the stance of a tweet toward its subject. We classify it as negative, neutral, or positive. Negative and positive tone reflect unfavorable and favorable language, respectively, whereas neutral tone captures descriptive statements without evaluative content. Our manual annotation is designed to capture nuances that dictionary-based methods often miss.⁴

Combining causality and tone yields a synthetic measure of rhetorical style. Tweets that are both causal and negative are coded as blame, while tweets that are both causal and positive are coded as credit. All remaining tweets are coded as none. Table A2 reports labeled examples.

3.2 Classification

To implement our measure of rhetorical style at scale, we begin by constructing a labeled dataset of 3,958 tweets, each annotated along the two dimensions introduced above: causality and tone. Because blame and credit tweets are likely to represent only a minority of the full corpus, a purely random sample would yield too few examples of the classes most relevant for training. This matters especially in our setting because

⁴For instance, “Our policies avoided a tragedy” may register as negative sentiment in a dictionary-based approach, but it has positive tone under our coding.

fine-tuning performance depends on exposing the model to a sufficiently balanced set of examples across classes. To improve learning, we therefore combine random sampling with targeted oversampling, a common strategy in the classification literature (e.g., Davidson et al., 2017; He and Garcia, 2009; Talat and Hovy, 2016). Specifically, we first generate benchmark examples of blame and credit tweets for Democrats and Republicans using ChatGPT 4o, solely to guide sampling. We then compute cosine similarity between these benchmark examples and the full tweet corpus using SBERT-mini embeddings (Reimers and Gurevych, 2019). Half of the tweets selected for annotation are drawn from those most similar to the benchmarks, balancing by party and rhetorical style, while the remaining half are sampled at random.

To assess annotation reliability, we select a subsample of 530 tweets and have each of them independently labeled by all three coders. For this jointly annotated subsample, labels for causality and tone are assigned by majority vote, with ties broken randomly. The resulting protocol delivers strong inter-annotator agreement on the rhetorical style measure: the average pairwise correlation is 0.73 and Fleiss’ Kappa is 0.64, typically interpreted as substantial agreement.⁵ Coders agree on rhetorical style in 67 percent of tweets, roughly six times the rate expected by chance.

We then fine-tune a RoBERTa-large model pre-trained on 154 million tweets (Loureiro et al., 2022).⁶ We split the labeled corpus into 80 percent training data (3,166 tweets) and 20 percent validation data (792 tweets).⁷

The fine-tuned classifier assigns each tweet a probability distribution over the three classes—blame, credit, and none—and labels the tweet according to the highest probability class. Ambiguous cases, in which the model assigns similar probabilities to multiple classes, are extremely rare.⁸ In the full corpus, 16 percent of tweets are classified as blame, 20 percent as credit, and the remainder as none. On the validation set, the model achieves an accuracy of 0.83, an F1-score of 0.84, and a Matthews Correlation Coefficient of 0.73, indicating strong performance by the standards of comparable

⁵Fleiss’ Kappa is a standard statistic for assessing agreement among more than two coders when observations are assigned to mutually exclusive categories. It is the natural multi-coder extension of Cohen’s Kappa. Unlike simple percent agreement, it adjusts for the amount of agreement that would arise mechanically by chance, and therefore provides a more informative measure of annotation reliability.

⁶Fine-tuning adapts a pre-trained language model to a specific downstream task by updating its parameters on a smaller labeled dataset. In our setting, this allows the model to learn how causal attributions and tone are expressed in political language.

⁷We train for 10 epochs and retain the epoch with the highest F1 score on the validation set. As robustness checks, we also fine-tune (i) a version of the model with half of its layers frozen and (ii) the well-know BERTweet model (Nguyen et al., 2020). Both alternatives perform slightly worse, but their outputs remain highly correlated with those of our baseline classifier, with Matthews Correlation Coefficients of 0.93 and 0.85, respectively.

⁸Figure A1 shows that, conditional on a predicted label, the associated probability distributions are strongly left-skewed and concentrated near 1, indicating that the classifier is rarely uncertain.

text-classification tasks.

Our analysis relies on two main variables: an indicator for blame, equal to one when a tweet is classified as blame and zero otherwise, and an indicator for credit, equal to one when a tweet is classified as credit and zero otherwise.

A natural concern is that our construction might miss a broader category of causal rhetoric that links political actions to outcomes without taking either a negative or a positive stance, and therefore extends beyond blame and credit. In practice, however, such neutral-toned causal tweets are exceedingly rare in our hand-labeled data, accounting for just 0.4 percent of observations. This suggests that, as qualitative observation of political discourse would also imply, politicians almost never use causal rhetoric in a neutral way. Instead, when they make such causal claims, they typically do so to shape the attribution process through claims of blame or credit. To further validate this point, we train a separate classifier using only the binary causal label from the annotated dataset and find that its output is highly correlated with an indicator for whether a tweet is classified as either blame or credit ($\rho = 0.9$). Taken together, these patterns confirm that causal rhetoric in politics consists almost entirely of assigning blame and claiming credit.

3.3 Validation

We next validate our measure in two ways. We begin with an external validation exercise, showing that our classifications align closely with independent annotation benchmarks. We then turn to a set of internal validation exercises, showing that blame and credit tweets exhibit linguistic and semantic patterns that match an intuitive understanding of these categories.

We start by comparing our blame and credit indicators with measures from the America’s Political Pulse dataset, which tracks politicians’ policy-attack and credit-claiming statements across multiple communication channels beginning in 2023 (Westwood and Lelkes, 2024).⁹ These categories provide a natural, though not identical, external benchmark for our own. Policy-attack closely parallels blame, insofar as both capture negative rhetoric directed at political actors or their actions, while credit-claiming parallels credit, insofar as both capture positive rhetoric that associates favorable outcomes or accomplishments with political actors. Since these categories do not

⁹In their codebook, Westwood and Lelkes (2024) define policy-attack as “communications about objecting to or raising concerns about a specific policy, law, or court ruling; using fact-based arguments even if critical or negative; avoiding emotional appeals, inflammatory language, claims of extremism, or personal attacks on individuals involved with the policy, including accusations of lying or withholding information,” and credit-claiming as “communications about creating or passing legislation; securing government spending, grants, or funding; or emphasizing personal or party accomplishments in office.”

fully coincide, one would not expect perfect correspondence. Still, their close conceptual similarity suggests that the two sets of measures should align strongly. For our validation exercise, we restrict that dataset to tweets, so that the comparison is based on the same communication channel as our own measure; this yields 158,745 tweets through the end of our sample period.

We assess this alignment at two levels. First, we validate at the tweet level by comparing, for each tweet we can directly match across the two datasets, our label with the corresponding America’s Political Pulse classification. Over the common sample period, we obtain 101,689 matched tweets. The alignment is strong. For blame, our label agrees with the corresponding policy-attack label in 87 percent of matched tweets, and the correlation between the two binary labels is 0.62. For credit, agreement with credit-claiming is 84 percent, and the corresponding correlation is 0.60.

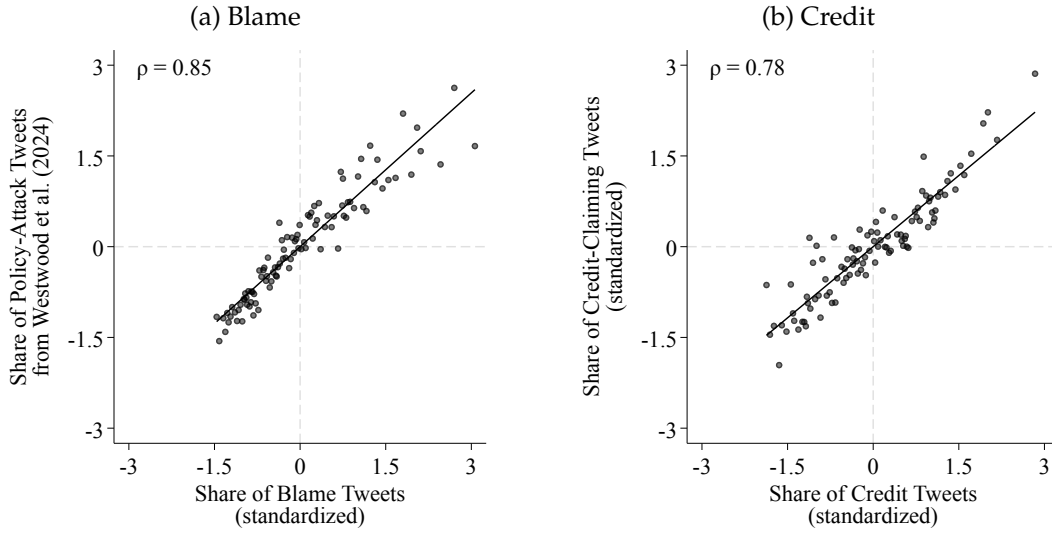
Second, we validate at the politician level. This is useful because several of our empirical results are about how intensively politicians use blame and credit, making it important to verify that our measure also aligns with external classifications in the cross-section of politicians. We therefore restrict attention to the 443 politicians observed in both datasets over the common sample period, from the beginning of 2023 through the end of our sample. For each of them, we compute the shares of policy-attack and credit-claiming tweets and correlate them with the corresponding shares of blame and credit tweets. Figure 1 shows strong positive correlations—0.85 between blame and policy-attack, and 0.78 between credit and credit-claiming. Taken together, the tweet-level and politician-level evidence provides strong external support for the validity of our measure.

We next turn to internal validation exercises, which help clarify the substantive content captured by our labels. We begin by examining the targets of causal attributions. Syntactically, we measure whether tweets are self- or other-referential by comparing the relative frequency of second- and third-person pronouns to first-person pronouns. Figure A2a shows that blame tweets are more outward-directed, framing responsibility around others, whereas credit tweets are more self-directed. Semantically, we classify whether the attribution targets Democrats or Republicans.¹⁰ Restricting attention to the 40 percent of tweets for which party identity can be identified, Figure A2b shows that blame is disproportionately directed at the opposing party, whereas credit is more often directed toward one’s own side.

We then examine the temporal orientation of causal rhetoric. Although our concept of causal rhetoric is not inherently temporal, blame and credit may differ in whether

¹⁰Target classification is based on the Political DEBATE language model developed by Burnham et al. (2025). Political DEBATE is specialized in zero-shot and few-shot classification of political documents, with performance on par with, or better than, state-of-the-art language models.

Figure 1: External Validation against America’s Political Pulse



Notes: This figure presents binned scatterplots comparing our politician-level measures of blame and credit with the corresponding politician-level measures from America’s Political Pulse (Westwood and Lelkes, 2024). In Panel (a), the horizontal axis reports the share of tweets classified as blame in our data, while the vertical axis reports the share of tweets classified as policy-attack in America’s Political Pulse. Panel (b) is analogous for credit and credit-claiming. Both sources are restricted to the common sample period, from the beginning of 2023 through the end of our sample. All variables are standardized. Observations correspond to politicians and are grouped into 100 bins. The correlation coefficient is reported in the upper-left corner of each panel.

they connect actions to realized outcomes or to expected consequences. We capture this distinction using normalized counts of past and future tense. Figure A2c shows that blame is predominantly retrospective, whereas credit is more forward-looking, consistent with findings from psychology (Malle et al., 2014).

We next study the emotional content associated with each rhetorical style. Using a RoBERTa model fine-tuned for emotion detection in tweets (Camacho-Collados et al., 2022), we decompose the emotional profile of blame and credit tweets. Figure A3 shows that blame is dominated by anger (nearly 50 percent), followed by disgust and fear, consistent with the idea that anger is directional and closely tied to causal attribution (Lazarus, 1991). Credit, by contrast, is overwhelmingly associated with optimism (over 50 percent), reinforcing the earlier result that credit is more forward-looking.

Finally, we examine diagnostic language. Following Gentzkow and Shapiro (2010), we extract the bigrams that are most distinctive of each category. As shown in Figures A4 and A5, these bigrams closely match both intuition and the linguistic patterns documented above.¹¹ We complement these diagnostics with illustrative tweets in Tables A3 and A4.

¹¹For Democrats, “trump administration” is diagnostic of blame, while “will help” is diagnostic of credit. For Republicans, “southern border” characterizes blame, while “will help” again identifies credit.

3.4 Distinctiveness Relative to Existing Text Measures

To conclude, we show that our measures do not collapse onto a combination of well-established linguistic features of political language. Specifically, we compare blame and credit with three prominent dimensions studied in the literature: sentiment, emotionality, and moral terminology.¹² We estimate tweet-level regressions of the form:

$$y_i = \alpha + \beta_1 \text{Sentiment}_i + \beta_2 \text{Emotionality}_i + \beta_3 \text{MoralValues}_i + \varepsilon_i, \quad (1)$$

where $y_i \in \{\text{Blame}_i, \text{Credit}_i\}$ is a binary indicator for whether tweet i is classified as blame or credit, Sentiment_i denotes sentiment, Emotionality_i denotes emotionality, and MoralValues_i denotes moral terminology. All regressors are standardized, and errors are clustered at the politician level.

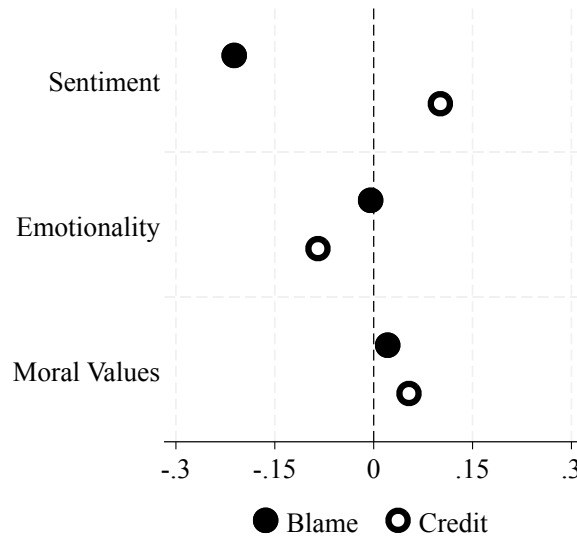
Figure 2 shows that blame and credit are correlated with existing linguistic dimensions in intuitive ways, but are not reducible to them. The estimated coefficients remain modest in magnitude, reaching at most about 0.2 in absolute value. This suggests that blame and credit capture a distinct dimension of political discourse rather than simply relabeling existing textual features. At the same time, these modest correlations are informative and provide an additional validation exercise. Sentiment is the strongest correlate of both outcomes: blame is negatively associated with sentiment, whereas credit is positively associated with it. This pattern is expected, given that blame and credit partly inherit the evaluative direction of the underlying language. Both blame and credit are negatively associated with emotionality, supporting the view that causal rhetoric is expressed in relatively more reasoning-oriented terms. This relationship is much stronger for credit than for blame, which may reflect a higher evidentiary burden when claiming credit than when assigning blame. Finally, both dimensions are only weakly positively associated with moral terminology.

4 The Supply of Causal Rhetoric

Relying on our measure of blame and credit, we show that causal rhetoric has become increasingly central in politicians' communication on social media, focusing on

¹²For sentiment, we use the [Camacho-Collados et al. \(2022\)](#) model, which assigns to each tweet a probability of being negative, neutral, or positive. We convert these probabilities into a continuous sentiment measure defined as $-1 \times \text{Prob}(\text{negative}) + 0 \times \text{Prob}(\text{neutral}) + 1 \times \text{Prob}(\text{positive})$. To measure emotionality and moral terminology, we follow the procedures in [Gennaro and Ash \(2022\)](#) and [Enke \(2020\)](#), respectively. For emotionality, we embed emotional and reasoning words and compute, for each tweet, the ratio of cosine similarities with the emotional versus reasoning embedding. For moral terminology, we compute, for each moral value, the average of vice and virtue frequencies, sum across all moral values, and normalize by the number of non-stop words.

Figure 2: Distinctiveness Relative to Existing Text Measures



Notes: The figure presents the estimates of Equation 1. Bars represent 95 percent confidence intervals computed with standard errors clustered at the politician level.

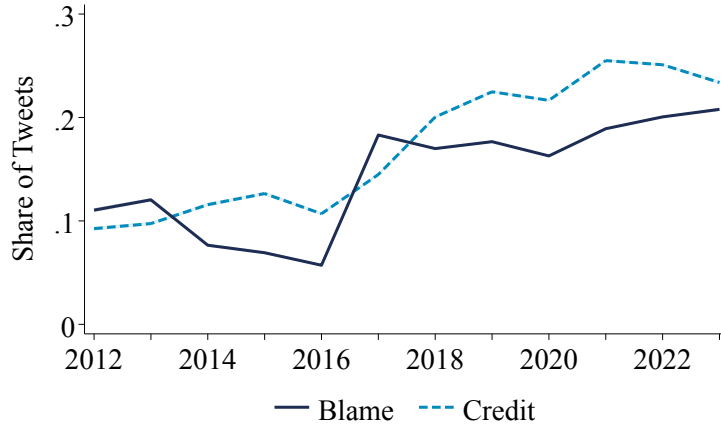
the case of congressional tweets. We document three main facts. First, the share of both blame and credit tweets roughly doubles over the sample period, with most of the increase concentrated between 2017 and 2020. Second, this rise occurs within politicians and within topics, indicating that the increased supply of causal rhetoric is not driven by compositional changes, but reflects choices made by politicians in how they communicate on social media. Third, causal rhetoric increasingly substitutes for purely affective messaging—tweets expressing positive or negative sentiment without causal claims—and becomes a central channel through which elite polarization is expressed.

4.1 Blame and Credit over Time

Figure 3 plots the yearly share of blame and credit tweets from 2012 to 2023. Both dimensions rise substantially over the sample period. Both increase increases from about 10 percent to more than 20 percent. Taken together, their share grows from roughly one fifth to nearly one half of all congressional tweets, indicating that causal rhetoric has become a prominent feature of political communication on social media.

The increase is also steep. The share of blame tweets rises by more than 10 percentage points in a single year, between 2016 and 2017. Credit follows a similar trajectory, although its growth is more gradual and concentrated between 2017 and 2020. By 2020, both series appear to plateau at high levels, suggesting that blame and credit became entrenched components of congressional communication. Overall, these patterns show that causal rhetoric became much more central in politicians social media communication, with the 2017–2020 period marking the key phase of this expansion.

Figure 3: Blame and Credit over Time



Notes: The figure presents the yearly share of tweets classified as blame and credit. Shaded areas represent 95 percent confidence intervals.

We probe the robustness of these facts in several ways. First, Figure A6 replicates the analysis excluding Senators’ tweets, which are unavailable prior to 2017, and yields very similar patterns. Second, Figure A7 shows that no other textual feature exhibits a comparable trend. Finally, in Appendix B.1 we show that the rise of causal communication appears unrelated to changes in Twitter’s policies.

4.2 A Pervasive Increase in the Supply of Causal Rhetoric

The time trends above show a clear increase in the supply of causal rhetoric, but by themselves they do not establish how pervasive that increase is. In principle, the rise in blame and credit could simply reflect compositional changes, driven either by turnover in congressional membership or by a growing focus on topics that are naturally more prone to causal rhetoric. To assess this possibility, we estimate an event-study specification centered on 2012 and progressively introduce politician and topic fixed effects.

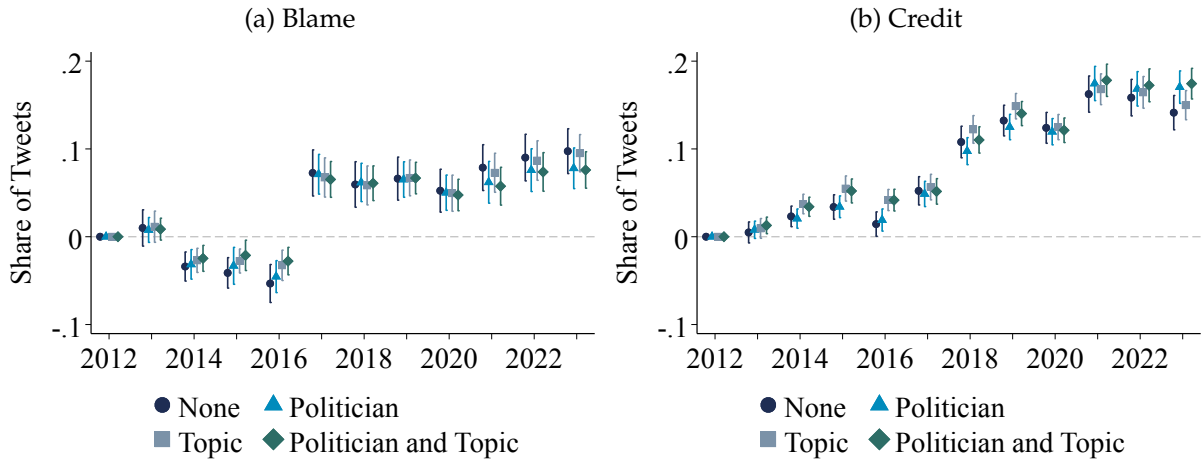
We classify each tweet into one of eight broad topics—economy, environment, healthcare, and immigration on the policy side; gender, gun control, policing, and racial relations on the sociocultural side—and estimate the following specification:¹³

$$y_{iptj} = \sum_{k=2013}^{2023} \beta^k \mathbb{1}[j = k] + \lambda_p + \mu_t + \varepsilon_{iptj}, \quad (2)$$

where y_{iptj} is an indicator equal to one if tweet i , posted by politician p on topic t in year j , is classified as blame (or credit), and zero otherwise; λ_p and μ_t denote politician

¹³Topic classification also relies on the Political DEBATE language model by Burnham et al. (2025).

Figure 4: A Pervasive Increase in the Supply of Causal Rhetoric



Notes: The figures present the estimated β^k from Equation 2. Each color corresponds to a specification including the fixed effects indicated in the legend. Bars represent 95 percent confidence intervals computed with standard errors clustered at the politician level.

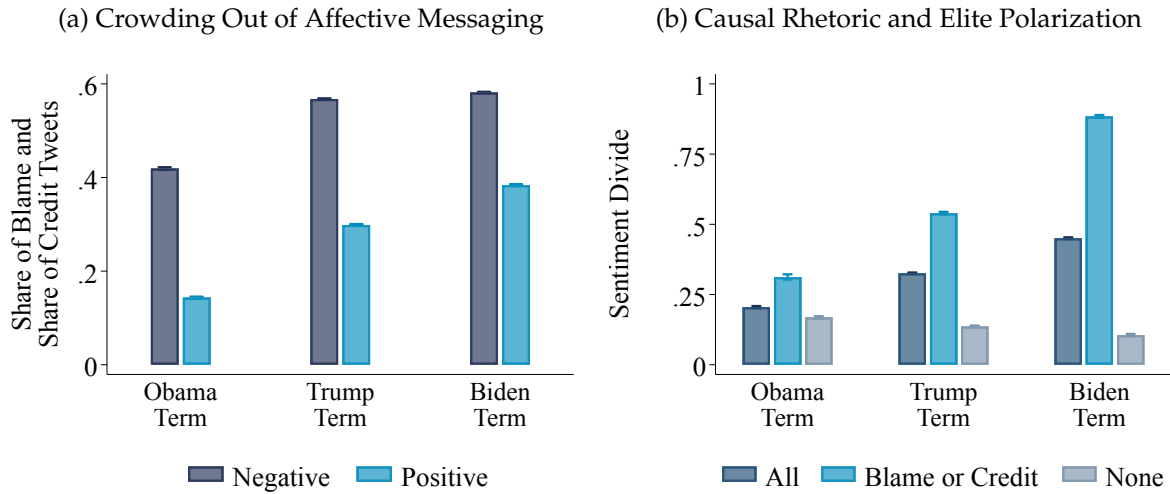
and topic fixed effects.

Figure 4 reports the estimated coefficients as fixed effects are added. The results show that the rise in causal rhetoric is not driven by composition. Blame increases by about 10 percentage points between 2012 and 2023 in the raw data, and by 7.6 percentage points even after controlling for both politician and topic fixed effects. For credit, the increase is even larger: the unconditional rise is about 14 percentage points, and it expands to 17 percentage points once composition is accounted for.

To complete the picture, we show that this rise is relatively homogeneous across politicians but quite heterogeneous across topics. On the politician side, Figure A8 shows that demographic characteristics have little explanatory power for whether a tweet is classified as blame or credit, with ideology as the main exception: more extreme politicians are more likely to use blame, whereas more moderate politicians are more likely to use credit. Consistently, Figure A9 shows that, among members serving both before and after 2016, within-politician changes are broadly positive and display only moderate dispersion. On the topic side, by contrast, Figure A10 shows that causal rhetoric is concentrated in policy domains rather than sociocultural ones, while Figure A11 shows that most of the increase over time is driven by policy topics.

Taken together, these results show that the rise of causal rhetoric is not a mechanical consequence of who is speaking or what is being discussed. Rather, it reflects a broad within-politician and within-topic increase in the supply of causal rhetoric, consistent with politicians making greater use of these rhetorical tools.

Figure 5: Causal Rhetoric, Affective Messaging, and Polarization



Notes: Panel (a) reports the share of blame tweets among tweets with negative sentiment and the share of credit tweets among tweets with positive sentiment, separately by presidency. Panel (b) reports the difference in average standardized sentiment between tweets posted by members of the ruling party and those posted by members of the opposition, separately by presidency. We compute this difference for three sets of tweets: all tweets, tweets classified as blame or credit, and non-causal tweets. In both panels, bars represent 95 percent confidence intervals.

4.3 Causal Rhetoric, Affective Messaging, and Polarization

As causal rhetoric becomes more central in politicians' communication, two related questions arise. First, what type of messaging does it replace? Second, how does its rise reshape the way polarization is expressed among elites? The evidence suggests that causal rhetoric increasingly crowds out purely affective messaging—that is, tweets that express positive or negative sentiment without causal claims—and, in doing so, becomes a central vehicle of elite polarization.

Figure 5a plots the share of blame tweets among those expressing negative sentiment and the share of credit tweets among those expressing positive sentiment, separately by presidency. The pattern is clear: blame increasingly occupies the negative sentiment space, rising from 42% under Obama to 58% under Biden, while credit increasingly occupies the positive sentiment space, rising from 14% to 38%.

This change may matter not only for the composition of political messaging, but also for how partisan conflict is conveyed. We therefore ask whether the rise of causal rhetoric has reshaped the expression of affective polarization among elites. Figure 5b shows that it has. While the overall sentiment gap between the ruling party and the opposition widens over time, causal tweets entirely drive this increase. By the Biden presidency, sentiment differences in non-causal tweets are close to zero, whereas causal tweets display a divide approaching 0.9.¹⁴ Taken together, these findings sug-

¹⁴A natural concern is that this pattern may arise mechanically because blame and credit tweets, by construction, refer to political actors and therefore may be more likely to contain partisan slant,

gest that causal rhetoric not only crowds out purely affective messaging, but also becomes the primary vehicle through which elite polarization is expressed.

5 The Impact on Small Donations

To study the supply of causal rhetoric in economic terms, we interpret its use through the lens of a production problem (Aridor et al., 2024), in which politicians choose rhetorical inputs based on both the returns they generate and their relative convenience in a given political context. A natural return of interest is political support: to what extent can politicians increase support from the electorate by assigning blame and claiming credit? Yet support is often difficult to observe, especially at the frequency at which politicians communicate on social media. To overcome this issue, we turn to campaign contributions from small donors, which can provide a high-frequency revealed-preference proxy of political support. Indeed, although small donations often account for only a limited share of total campaign revenue, they can be more broadly interpreted as a measure of electoral support and political mobilization (Bouton et al., 2022; Petrova et al., 2021). Small contributions first capture the total amount of money politicians are able to raise from their supporters. At the same time, donation data also allow us to disentangle two distinct margins of political support. On the one hand, by focusing on the number of donors, we can assess the extent to which causal rhetoric mobilizes support. On the other hand, by focusing on the average amount contributed per donor, we can assess the extent to which it deepens support.

Leveraging quasi-exogenous geographical variation in Twitter adoption (Müller and Schwarz, 2023), we show that causal rhetoric—especially blame—increases aggregate small-donor support. A decomposition reveals that the two dimensions operate through different margins: blame alone works at the extensive margin, by increasing the number of donors, whereas credit works only at the intensive margin, by increasing the average amount per donor. These effects also vary across donor types, with ideologically extreme donors responding more to blame and ideologically moderate donors responding more to credit. Finally, consistent with the broader rise of causal rhetoric, these patterns become stronger over time.

whereas non-causal tweets may less often target identifiable political actors. To alleviate this concern, Figure A12 shows that the results are unchanged when we restrict the analysis to tweets that can be clearly identified as targeting either Democrats or Republicans.

5.1 Empirical Strategy

To identify the causal effect of blame and credit on donations, we exploit cross-county variation in Twitter penetration across the United States. Intuitively, in counties with higher Twitter use, residents are more likely to be exposed to the blame and credit rhetoric supplied by politicians. We therefore estimate, for each politician, how monthly donations from a given county respond to that politician’s monthly level of blame and credit as a function of local Twitter penetration.

Empirically, we use the geographic detail in campaign finance records to construct a politician-by-county-by-month panel that tracks, for each politician, the total revenue from small donors, the number of small donors, and the average amount contributed per donor in each county-month in which the politician tweets. We then estimate the following specification:

$$y_{icm} = \beta_1(\text{Blame}_{im} \times \text{Users}_c) + \beta_2(\text{Credit}_{im} \times \text{Users}_c) + \gamma X_{icm} + \lambda_{ic} + \mu_{im} + \eta_{mc} + \varepsilon_{icm}. \quad (3)$$

Here, y_{icm} is the outcome of interest for politician i in county c and month m : $\log(1 + \text{revenue from donations})$, $\log(1 + \text{number of donors})$, or $\log(1 + \text{average donation per donor})$. Blame_{im} and Credit_{im} denote the shares of blame and credit tweets posted by politician i in month m , and Users_c measures Twitter penetration in county c . The term X_{icm} denotes a vector of controls. In our preferred specification, it includes only $\text{Volume}_{im} \times \text{Users}_c$, where Volume_{im} captures total tweet volume in month m . This allows us to separate the effect of rhetorical style from the broader effect of a politician’s overall social media activity as reflected in their tweet volume.

The fixed effects absorb a broad set of remaining confounders. Politician-by-month fixed effects (μ_{im}) absorb any month-specific shock to politician i , such as changes in overall popularity or fundraising, that is common across counties. Politician-by-county fixed effects (λ_{ic}) absorb persistent differences in politician i ’s support across counties, including particular geographic ties between politicians and specific counties. County-by-month fixed effects (η_{mc}) absorb any county-level shock to donations in month m that is common across politicians, such as local events or other time-varying factors affecting political engagement in that county. Thus, our coefficients are estimated from within-politician-month differences across counties with different levels of Twitter penetration, net of common county-month shocks and persistent politician-county links.

A central concern is that Twitter penetration may itself be endogenous to local political and economic conditions. Counties with greater partisan competition, higher

donor capacity, or different unobservable characteristics may exhibit different level of Twitter penetration and may also differ systematically in political engagement. If so, our estimates could confound the causal effect of rhetoric with differences in the demand for political communication. To address this concern, we exploit quasi-exogenous variation in Twitter penetration induced by the platform’s early diffusion after the South-by-Southwest (SXSW) festival in 2007, a well-documented shock that accelerated Twitter’s initial spread in the United States (Fujiwara et al., 2024; Müller and Schwarz, 2023). Specifically, we instrument county-level Twitter penetration with the number of followers of the official SXSW Twitter account who joined Twitter in 2007. This instrument provides plausibly exogenous variation in local Twitter use, orthogonal to local determinants of political donations.¹⁵

In practice, we instrument $Users_c$ with $\log(1 + SXSWFollowers2007_c)$, where $SXSWFollowers2007_c$ denotes the number of followers of the official SXSW account in county c who joined Twitter in 2007, after the SXSW Festival. We then estimate a standard two-stage least squares specification, instrumenting each interaction in Equation 3 that involves county Twitter penetration with the corresponding interaction based on $SXSWFollowers2007_c$. Following Fujiwara et al. (2024); Müller and Schwarz (2023), we also interact our main regressors of interest with the number of SXSW followers who joined Twitter before 2007, at any point in 2006. This ensures that identification comes from comparing counties that are similar in their pre-existing propensity to engage with SXSW, and therefore more comparable in their baseline relationship to the platform. Thus, the identifying assumption resembles a parallel-trends condition in spirit: conditional on the fixed effects, tweet-volume exposure, and pre-2007 SXSW propensity controls, counties more exposed to SXSW-induced Twitter adoption would have experienced similar changes in politician-specific donations across high-blame and low-blame, or high-credit and low-credit, months absent differential exposure to politicians’ tweets.

In addition, we estimate two richer specifications, each designed to address a distinct concern. First, one may worry that the coefficients on blame and credit simply capture the role of negative and positive sentiment, rather than the specific effect of causal attribution. This concern is natural because the omitted category in the baseline specification consists of tweets classified as none, so blame and credit are identified relative to tweets that differ both in causal content and in tone. To address this, we control for the shares of non-causal tweets with negative sentiment and of non-causal tweets with positive sentiment, each interacted with county Twitter penetration. This

¹⁵This instrument is now standard in the literature (e.g., Boken et al., 2023). We refer to the original papers for detailed evidence on both its relevance for the diffusion of local Twitter activity and its plausible exogeneity.

allows us to estimate the impact of blame and credit net of broader negative and positive tone, and therefore assess whether the returns we document are specific to the causal margin.

Second, despite the quasi-exogenous variation in Twitter penetration and the demanding fixed-effects structure, a residual concern is that blame and credit may co-move with other politically salient content. Politicians may assign more blame or claim more credit when economic news is released, scandals emerge, or other salient information becomes public. If such accompanying content independently affects donations and is itself amplified through Twitter, our estimates could partly capture the effect of that correlated content rather than the rhetorical margin itself. To alleviate this concern, we estimate a richer specification that allows the effects of blame, credit, and tweet volume to vary flexibly with county characteristics, using the same county-level covariates employed by [Fujiwara et al. \(2024\)](#). Because the baseline model already includes a demanding set of fixed effects, we address the resulting dimensionality problem by summarizing 28 county-level covariates into seven principal components and interacting those components with blame, credit, and tweet volume.¹⁶ These covariates include not only demographic and socioeconomic characteristics, but also measures of media exposure such as prime-time television viewing and Fox News viewership. This matters because salient political events can spread through multiple channels, and many alternative explanations would predict stronger donation responses in counties with different media-consumption patterns, demographic compositions, urbanicity, or general responsiveness to political news. By allowing the effects of blame, credit, and tweet volume to vary along these county-level dimensions, the richer specification absorbs broad differences in how salient political content diffuses and translates into donations across county types. Accordingly, this additional specification narrows the identifying assumption: correlated political events must not be differentially amplified through the Twitter-exposure margin isolated by the instrument, over and above the broader observable media, demographic, socioeconomic, urbanicity, and political-news channels summarized by the principal components. Hence, this helps address the concern that the baseline estimates capture correlated events rather than the rhetorical margin itself.

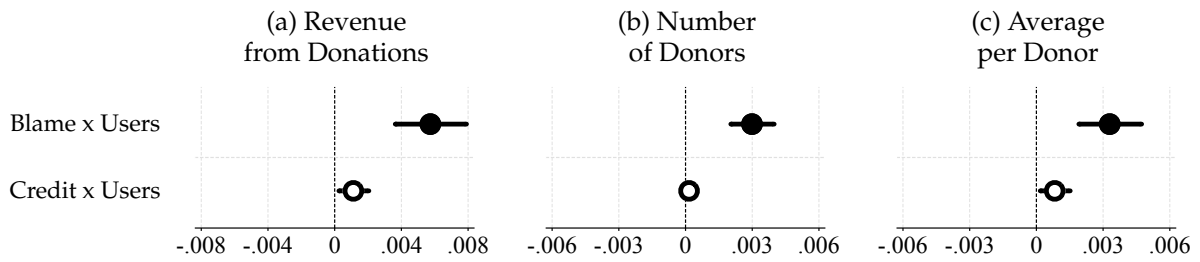
5.2 Results: Aggregate Returns and Their Margins

We begin by estimating Equation 3 using total revenue from small donations as the outcome. Results are presented in Figure 6a.¹⁷ We find that blame has a positive

¹⁶Table A5 lists the 28 underlying county-level covariates.

¹⁷Table A6 reports the main 2SLS estimates. Table A7, Table A8, and Table A9 report the corresponding OLS, reduced-form, and first-stage results.

Figure 6: Aggregate Returns and Their Margins



Notes: The figure presents the estimates of Equation 3, considering as outcome log-donation revenue, log-number of donors, and log-average donation per donor in Panels (a), (b), and (c), respectively. Bars represent 95 percent confidence intervals computed with standard errors clustered at the county level.

and precisely estimated effect on donation revenue, while credit has a smaller but still positive effect. In particular, in our preferred specification, a one-standard-deviation increase in the monthly share of blame tweets raises donation revenue by roughly 3.2 percent in a county with average Twitter penetration, whereas the corresponding increase in credit raises revenue by about 0.5 percent. Both rhetorical styles therefore generate political returns, but blame is substantially more effective in the aggregate.

We next ask through which margin these returns accrue. Donation revenue can rise either at the extensive margin, through donor mobilization because a politician attracts more donors, or at the intensive margin, through support deepening because existing donors give more. To distinguish between these channels, we re-estimate the same specification using, in turn, the number of donors and the average amount donated per donor as separate outcomes.

The decomposition of donation revenue points to distinct political returns from blame and credit. Notably, blame is the only rhetorical style that manage to expand support through donor mobilization, whereas credit primarily works only by deepening support among those already willing to give. In our preferred specification, a one-standard-deviation increase in blame raises the number of donors by about 1.6 percent in a county with average Twitter penetration, while credit has essentially no effect on this margin. By contrast, a one-standard-deviation increase in credit raises the average donation per donor by about 0.5 percent in a county with average Twitter penetration. Blame also increases average donation size, indicating that its effects are not confined to the extensive margin. Taken together, the results suggest that blame operates on both margins of support, but is alone effective at mobilizing additional donors, whereas credit only deepens support.¹⁸

¹⁸Appendix B.2 shows that this mobilization pattern possibly extends beyond donations. Using county-level protest data and the same identification strategy, we find that blame—but not credit—also increases the incidence protest activity, consistent with the view that blame alone is effective at activating broader forms of political mobilization.

A natural concern is that these results rely on log transformations in the presence of many zeros (Chen and Roth, 2024). To address this, we re-estimate the same specification using as outcome a binary indicator equal to one if a politician receives at least one donation from a given county in a given month, and zero otherwise. This provides an even coarser extensive-margin measure of support. As shown in Table A10, the results remain consistent with the donor-count evidence: blame increases the probability of receiving any donation, whereas credit does not generate a comparable effect.

The baseline estimates are also robust to the two richer specifications discussed above, as shown in columns 3 and 4 of Table A6. First, when we control for non-causal negative and positive tweets interacted with county Twitter penetration, the coefficients on blame and credit remain stable. This indicates that the effects we document are not simply driven by broader sentiment, but are specific to the causal-attribution margin. Second, the estimates remain similar when we allow the effects of blame, credit, and tweet volume to vary flexibly with county characteristics through interactions with principal components of a rich set of county-level covariates. This alleviates the concern that the baseline estimates reflect correlated political events or other salient content that diffuses differently across county types.

To benchmark the magnitude of the mobilization effect, we compute persuasion rates following DellaVigna and Gentzkow (2010), focusing on the donor-count outcome.¹⁹ The implied persuasion rate is 0.94%.²⁰ This is comparable to estimates for political advertising (Spenkuch and Toniatti, 2018) and opening a Twitter account (Petrova et al., 2021), though smaller than the effects associated with virality (Boken et al., 2023) and the average persuasion rates summarized by DellaVigna and Gentzkow (2010).

Overall, the evidence points to a nuanced pattern of returns. Causal rhetoric—especially blame—increases small-donor support, but different rhetorical styles do so through different channels. Blame is the only driver of donor mobilization, while credit works by increasing only the intensity of support conditional on giving. In the remainder of the section, we probe these mechanisms further by studying heterogeneity across donors, the virality of blame, and the evolution of these effects over time.

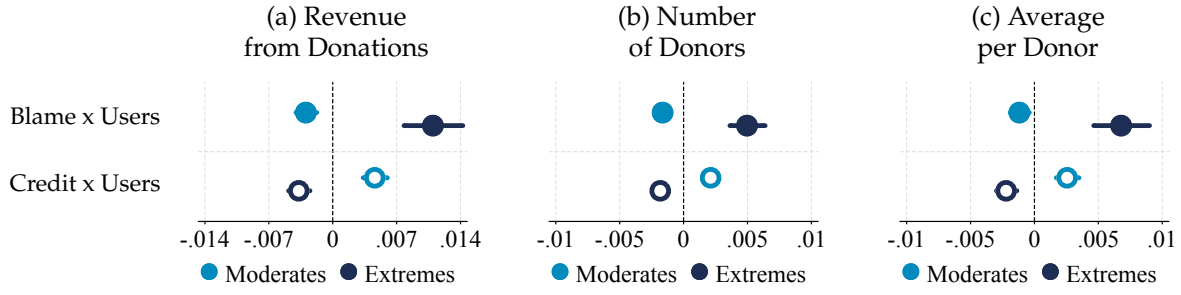
5.3 Heterogeneity by Donor Ideology

A further layer of nuance emerges once we allow returns to vary across donor types. While the previous results show that blame and credit differ in how they trans-

¹⁹As emphasized by Boken et al. (2023), persuasion rates are most naturally interpreted when the underlying outcome is close to a binary participation decision.

²⁰Appendix B.3 details the construction of this estimate and shows that it should be interpreted as a lower bound.

Figure 7: Heterogeneity by Donor Ideology



Notes: The figure presents the estimates of Equation 3, considering as outcome log-donation revenue, log-number of donors, and log-average donation per donor in Panels (a), (b), and (c), respectively. Each outcome is computed separately for ideologically moderate and extreme donors. Bars represent 95 percent confidence intervals computed with standard errors clustered at the county level.

late into aggregate support, they may also differ in whose support they attract.

To study this heterogeneity, we use the donor-level ideology measure constructed by Bonica (2014) from campaign finance records, analogous to Nominate scores for legislators. Within each electoral cycle, we classify donors as moderate or extreme depending on whether the absolute value of their ideological score falls below or above the cycle-specific median. We then re-estimate Equation 3 separately by donor type.

Figure 7 reveals a clear ideological divide.²¹ Moderate donors respond positively to credit and negatively to blame, whereas extreme donors display the opposite pattern: they respond positively to blame and negatively to credit. The symmetry of this pattern is apparent, but it is not quantitatively balanced: the positive response of extreme donors to blame is substantially stronger than the positive response of moderate donors to credit. These differences suggest that blame and credit do not merely operate through different margins of support; they also appeal to different segments of the donor base. In this sense, the two rhetorical styles are politically complementary not only because they mobilize support in different ways, but also because they resonate with different ideological constituencies.

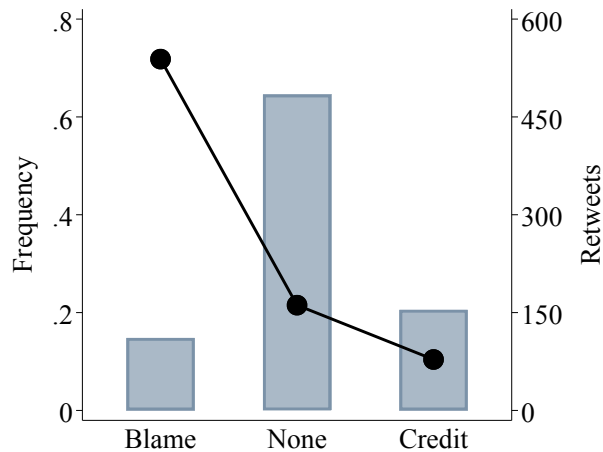
5.4 Mechanism: Blame and Virality

A natural mechanism for the extensive-margin effect of blame is reach. If blame mobilizes support by bringing in additional donors, one may expect blame tweets to spread more widely across the platform. We therefore examine the relationship between rhetorical style and retweet activity. Appendix B.4 provides details on the data and empirical specifications.

Figure 8 describes the relationship between rhetorical style and retweet activity.

²¹Tables A11 and A12 report the corresponding estimates.

Figure 8: Blame and Virality



Notes: The figure compares the frequency and average retweet count of tweets classified as blame, none, and credit, restricting to tweets with non-missing retweet data. The bars report the share of tweets in each rhetorical category (left axis), while the connected line reports the average number of retweets received by tweets in that category (right axis).

Although blame tweets represents a minority of the corpus for which retweet data are available, they receive on average nearly four times as many retweets as other tweets. A back-of-the-envelope calculation underscores the magnitude of this pattern: while blame represents only 15% of tweets, it generates nearly 40% of all retweets.

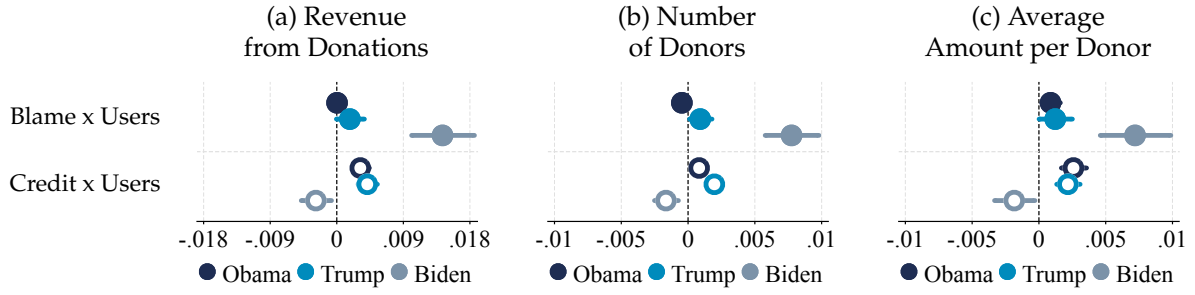
This pattern also emerges in more structured analyses. Relative to non-causal tweets, blame tweets receive about 0.2 standard deviations more retweets even after controlling for politician, year, and topic fixed effects. Moreover, this effect is not uniform across the retweet distribution. Blame is negatively associated with appearing in the lower deciles of retweet counts, but increasingly positively associated with the upper deciles, with the strongest effect in the top decile. In other words, blame is not simply more engaging on average; it is disproportionately more likely to generate highly viral content. The association between blame and virality also strengthens over time, mirroring both the rise in the supply of blame and its growing returns in terms of donations. By contrast, credit displays no comparable pattern.

Taken together, these findings support the interpretation that blame mobilizes support in part because it travels farther. In this sense, virality provides a plausible mechanism linking blame rhetoric to the increase in donor participation documented above, and connects our evidence to [Boken et al. \(2023\)](#), who show that small donations rise sharply when politicians go viral.

5.5 Returns over Time

We conclude by examining whether the returns to causal rhetoric evolve alongside its changing supply. Although this exercise is mainly descriptive, it helps clarify

Figure 9: Returns over Time



Notes: The figure presents the estimates of Equation 3, considering as outcome log-donation revenue, log-number of donors, and log-average donation per donor in Panels (a), (b), and (c), respectively. The specifications allow the effects of blame and credit to vary across the three presidencies. Bars represent 95 percent confidence intervals computed with standard errors clustered at the county level.

whether the growing use of blame and credit by politicians is accompanied by changes in their political returns. To do so, we re-estimate Equation 3, allowing the coefficients on the interactions ($\text{Blame}_{im} \times \text{Users}_c$) and ($\text{Credit}_{im} \times \text{Users}_c$)—as well as the corresponding pre-SXSW control interactions—to vary by presidential administration.

Figure 9 reports the resulting estimates.²² The results reveal a marked increase in the returns to blame over time. Under Obama, blame has essentially no returns. Under Trump, its returns become positive, and under Biden they strengthen further, following a monotonic pattern. The pattern for credit is different. Credit generates positive returns under both Obama and Trump. Under Biden, however, these effects reverse sign: credit becomes negatively associated with all three donation outcomes. Thus, unlike blame, credit does not display a monotonic increase in returns over time.

Overall, these time patterns mirror the broader supply dynamics documented above. As blame becomes increasingly central in elite communication, its returns rise sharply as well. This is especially true during the Biden period, when blame is both most prevalent and most effective. Credit, by contrast, does not follow the same trajectory.

6 Strategic Blame and Credit

Having characterized the political returns to blame and credit, we now turn to their strategic use. The fact that these rhetorical styles generate returns does not, by itself, imply that politicians deploy them strategically. But if blame and credit are strategic rhetorical inputs, their use should reflect not only the returns they generate on average, but also their relative convenience in a given political context—just as the allocation of inputs in a production problem depends both on returns and on how easily each input can be used in a given environment. We argue that politicians reallocate blame

²²Table A13 reports the detailed estimates.

and credit strategically as the political environment changes. A key force shaping this allocation is credibility. Causal claims cannot be advanced arbitrarily: to convince, they must remain sufficiently tied to observable political reality.²³ Policy ownership shifts this credibility margin. When the opposing party owns political action, blame becomes more credible; when one’s own party does, credit becomes more credible.

We present two patterns in support of this argument. First, raw choice data show that politicians use both blame and credit, and that the allocation between them varies systematically by institutional position. Second, using bill introductions as a proxy for policy ownership, we show that blame rises when the opposing party acts, whereas credit rises when one’s own party does.

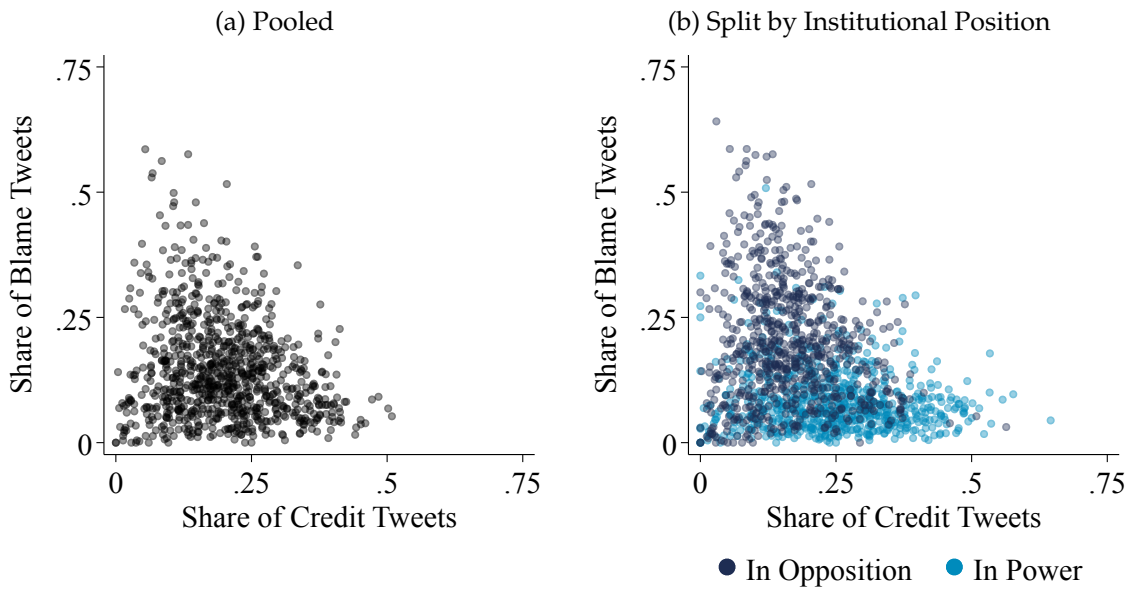
6.1 The Allocation of Blame and Credit

We begin with raw choice data to study how politicians allocate rhetoric between blame and credit. Figure 10a plots, for each member of Congress, the share of tweets classified as blame against the share classified as credit. Two facts stand out. First, politicians differ substantially in how they allocate rhetoric across the two styles. Second, they use both blame and credit, rather than concentrating exclusively on blame, even though the previous section showed that blame is, on average, more effective in generating political support. Taken together, these patterns suggest that politicians solve a non-trivial allocation problem, and that they solve it differently: they do not converge on a common rhetorical mix, nor do they resolve the problem with a uniform corner solution on the input with the highest average return. There are, of course, several possible reasons for this pattern. For instance, as we have shown, blame and credit appeal to different ideological audiences. Politicians may therefore face heterogeneous constituencies, or be uncertain about their composition, leading them to smooth between the two rhetorical styles. But a sharper pattern emerges once we reorganize the same allocation by institutional position.

Figure 10b plots again, for each member of Congress, the share of blame tweets against the share of credit tweets, but now separating politician-position observations according to whether the politician is in power or in opposition, defined by whether the politician’s party holds the U.S. presidency. In practice, each politician enters with a separate observation in each institutional position they occupy. The pattern is immediate: the allocation of blame and credit is systematically organized by whether politicians are in power or in opposition. Members in opposition rely relatively more on blame, whereas members in power rely relatively more on credit. This organization

²³This is in the same spirit as the theoretical narratives literature, where narratives are often required to remain at least minimally consistent with the evidence (Bilotta and Manferdini, 2026; Eliaz et al., 2025; Eliaz and Spiegler, 2020).

Figure 10: The Allocation of Blame and Credit



Notes: Panel (a) plots, for each member of Congress, the share of tweets classified as blame against the share classified as credit. Panel (b) plots the same relationship at the politician-position level, separating observations according to whether the politician is in power or in opposition, where institutional position is defined by whether the politician's party holds the U.S. presidency. Politicians or politician-position observations with fewer than 10 tweets are excluded.

by institutional position is difficult to reconcile with a pure audience-heterogeneity account. Rather, it fits the idea that politicians reallocate blame and credit strategically as the political environment changes, shifting toward the claim that is more credible. Because causal claims must remain tied to observable political reality to convince, policy ownership shifts which of the two rhetorics is easier to sustain. When the opposing party is responsible for political action, blame is more credible; when one's own party is responsible, credit is more credible.

6.2 Blame, Credit, and Policy Ownership

To investigate this credibility mechanism more directly, we move from institutional position to a more fine-grained proxy for policy ownership. Measuring policy ownership is inherently difficult, because responsibility for political action is often diffuse in democratic institutions. We therefore use bill introductions as observable instances in which policy ownership becomes comparatively clear. The implication is direct: if policy ownership shifts the relative credibility of blame and credit, politicians should be more likely to use blame when the opposing party acts and more likely to use credit when their own party acts.

We estimate two versions of the following specification:

$$y_{ipt} = \beta_1 \text{Opposing}_t + \beta_2 \text{Own}_t + \lambda_p + \mu_{week} + \varepsilon_{ipt}. \quad (4)$$

Here, y_{ipt} is a binary indicator equal to one if tweet i posted by politician p on day t is classified as blame (or credit); Opposing_t is an indicator equal to one if the opposing party introduces a bill on day t ; and Own_t is defined analogously for the politician's own party. The specification includes politician fixed effects, λ_p , and week fixed effects, μ_{week} , so identification comes from within-politician and within-week changes in rhetorical style around days in which either party introduces bills.

Figure 11 summarizes the results.²⁴ The pattern is sharp. Blame rises when the opposing party introduces bills: blame tweets are about 5 percentage points more likely to be posted on those days. Credit displays the mirror image: it rises when the politician's own party introduces bills, with credit tweets about 6 percentage points more likely to be posted. Relative to the baseline probabilities of posting blame and credit tweets in the sample, these correspond to increases of about 31 and 30 percent, respectively. Politicians thus shift toward blame when the opposing party owns the action, and toward credit when their own party does.

A natural concern is that these patterns reflect generic shifts in partisan tone rather than a mechanism specific to causal rhetoric. To address this, in Table A15 we re-estimate the specification restricting the sample to negative tweets when the outcome is blame and to positive tweets when the outcome is credit. This means that the comparison is now within tone: for blame, we ask whether politicians become more likely to post causal negative tweets rather than other negative tweets, and analogously for credit within positive tweets. The same pattern remains. This suggests that the mechanism is specific to causal rhetoric, rather than a generic feature of partisan tone in political discourse.

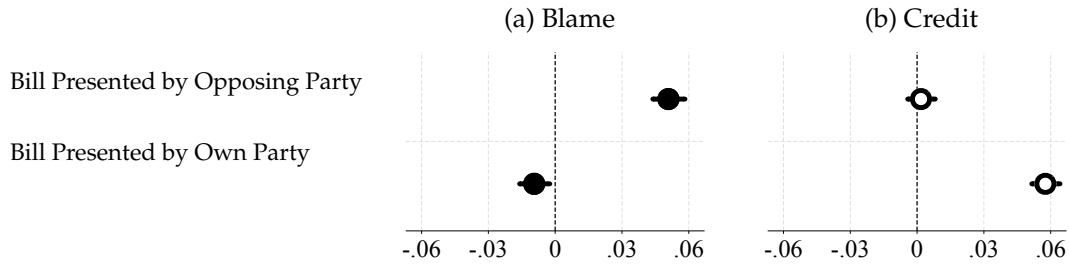
These patterns support the credibility mechanism proposed above. Bill introductions proxy for moments in which policy ownership becomes especially salient. The evidence therefore points to politicians strategically reallocating blame and credit as policy ownership changes which claim is more credible.

7 Assessing Economic Causal Rhetoric

We conclude by assessing the substantive grounding of the causal claims embedded in politicians' rhetoric. The previous sections showed that causal rhetoric is both politically effective and strategically deployed: it generates returns, especially through

²⁴Table A14 reports the detailed estimates.

Figure 11: Blame, Credit, and Policy Ownership



Notes: The figure presents the estimates of Equation 4, considering as outcomes a binary indicator for whether the tweet is blame and a binary indicator for whether the tweet is credit in Panels (a) and (b), respectively. Bars represent 95 percent confidence intervals computed with standard errors clustered at the day level.

blame, and politicians adjust its use to contexts in which it is more convenient or more credible. This leaves open a natural question: what, exactly, are donors being convinced of? If the claims conveyed by causal rhetoric are well grounded, then the patterns we document may reflect voters linking political actions to outcomes more accurately. If, instead, these claims rest on weak or heavily qualified foundations, then the same patterns reflect something different: causal rhetoric works not by improving how voters connect political actions to outcomes, but by compressing complex causal processes into politically convenient attributions. To assess this possibility, we focus on economic tweets and benchmark their dominant causal claims against mainstream macroeconomic knowledge. Two findings emerge. First, most economic causal claims are not cleanly supported as presented, but instead require qualifications. Second, this lack of grounding is especially pronounced for blame, which is also the rhetorical style that proves most politically effective.

7.1 Classification Procedure

Investigating this question is challenging. The object of interest is not a set of factual statements, but a set of causal claims. Unlike statements about realized inflation, spending amounts, or immigration levels, causal claims rarely admit a single observable benchmark against which they can be directly checked. The same claim may be true under some conditions but false under others, or supported in part yet fail once relevant qualifications are introduced. In other cases, we may simply not know whether the underlying causal link holds, and establishing it may itself require substantial research. In light of these difficulties, rather than aiming to determine whether each claim is true or false, we assess the extent to which it is supported by the current state of knowledge. By this, we mean the extent to which the claim is consistent with a disciplined body of accumulated knowledge in the relevant domain, including its standard mechanisms, qualifications, and areas of uncertainty.

Implementing this idea across the full set of tweets posted by politicians in our corpus would require comparing claims drawn from many different domains against the corresponding bodies of specialized knowledge, which would hardly be practically feasible. We therefore tackle this question by restricting the domain and, in particular, by focusing on economic tweets. This choice is guided by two considerations. First, causal rhetoric is especially prevalent in this domain: about 61 percent of tweets about the economy are classified as blame or credit, the highest share across topics. Second, claims in this domain can be assessed against a relatively well-defined body of mainstream macroeconomic knowledge and reasoning.

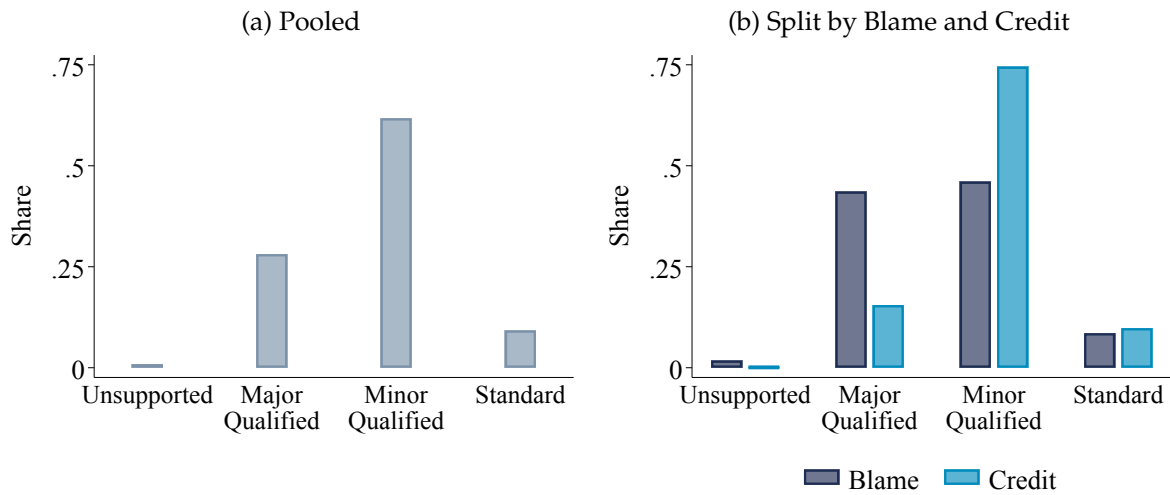
In practice, we analyze 413,836 blame and credit tweets that our topic classification identifies as being about the economy and evaluate the dominant causal claim in each tweet using a structured AI-assisted coding procedure.²⁵ For each tweet, we extract its main causal claim, identify its implied economic mechanism, and assign one of four labels. A claim is *supported* if it is backed by mainstream macroeconomics under ordinary assumptions, without any relevant additional qualification. It is *minor qualified* if it is supported by a standard mechanism and direction, but only once familiar caveats about context, timing, magnitude, or incidence are taken into account. It is *major qualified* if a route to support exists within mainstream macroeconomics, but only once substantial unstated conditions or offsetting forces are introduced. Finally, it is *unsupported* if no ordinary plausible mainstream macroeconomic route supports the claim under its baseline interpretation. In sum, this procedure asks how far these causal claims, as presented to voters, are supported by standard macroeconomic knowledge.

7.2 Support Levels

Figure 12a reports the distribution of labels across all economic tweets in our sample. The first result is that most economic causal claims are not cleanly grounded in standard macroeconomic reasoning: about 90% require either minor or major qualifications, while only about 10% are classified as supported. In other words, fewer than one in ten claims is supported in the form in which it is presented to voters. This pattern suggests that the persuasive content of causal rhetoric rarely consists of claims that standard macroeconomic knowledge would endorse without reservation. At the same time, the distribution of labels is informative also in another way. The largest mass of tweets lies in the qualified categories—62% minor qualified and 28% major qualified—rather than in the unsupported category, which accounts for less than 1% of observations. This points to politicians generally avoiding implausible or outlandish

²⁵We implement this procedure using GPT-5 mini under a fixed codebook and prompt that instruct the model to act as a careful research coder and evaluate claims using standard mainstream macroeconomic reasoning. We refer to Appendix B.5 for additional details.

Figure 12: Support Level for Economic Causal Rhetoric



Notes: Panel (a) reports the distribution of macroeconomic-support labels across all tweets that our topic classification identifies as being about the economy. Panel (b) reports the same distribution separately for tweets classified as blame and credit.

economic claims. Instead, they appear to operate in a region where the underlying mechanism often has some basis in standard macroeconomic reasoning, but only once qualifications are introduced that the rhetoric itself leaves unstated. Interpreted in this light, the evidence suggests that causal rhetoric often works by slanting the link between political actions and outcomes through the omission of qualifications, rather than by making claims that are plainly inconsistent with standard macroeconomic knowledge. All together, this raises a concern about what donors are being convinced of. Causal rhetoric, despite being politically effective, is rarely supported by standard macroeconomic knowledge as presented.

The results yield further insight once we distinguish between blame and credit. Figure 12b reports the distribution of labels separately for blame and credit tweets. It shows that blame is systematically less grounded in standard macroeconomic knowledge than credit. The sharpest contrast appears in the share of major qualified claims, which reaches 44% for blame but only 16% for credit. Credit, by contrast, is much more concentrated in the minor qualified category, accounting for 75% of claims, compared with 46% for blame. Unsupported claims remain rare for both rhetorical styles, yet they appear about ten times more often among blame tweets: 1.8% of blame claims are unsupported, compared with 0.17% of credit claims. Taken together, these comparisons show that blame lies farther from standard macroeconomic knowledge than credit. This finding is especially notable in light of our earlier results: blame is not only the least grounded form of causal rhetoric, but also the most politically effective. The concern, then, is not only that causal claims are highly qualified on average, but that the weakest grounding is concentrated precisely in the rhetoric that proves most

effective politically.

8 Conclusion

This paper studies causal rhetoric in politics: language that links political actions to outcomes through blame and credit. Using natural language processing, we measure causal rhetoric in a large corpus of congressional tweets and document that it has become increasingly central in politicians' communication. It rises pervasively over time, increasingly crowds out purely affective messaging, and becomes a main vehicle through which elite polarization is expressed.

We then study the economics of this rhetoric. We show that blame and credit both generate political returns, although blame is substantially more effective in the aggregate. We also find that the two operate through different margins and appeal to different constituencies: blame alone mobilizes additional donors and is especially effective among ideologically extreme donors, whereas credit only deepens support by increasing the average amount per donor and resonates more with moderates. We further find that politicians deploy causal rhetoric strategically. They reallocate between blame and credit as policy ownership shifts which of the two claims is more credible, relying more on blame when the opposing party owns the action and more on credit when their own party does. Finally, focusing on economic tweets and using an AI-assisted coding procedure, economic causal rhetoric appears to be only weakly grounded in mainstream macroeconomics, suggesting that it persuades less by clarifying the mapping from political actions to outcomes than by tilting those attributions in politicians' favor.

Taken together, our results also suggest several directions for future work. On the one hand, it would be useful to better understand why blame is more effective than credit, and through which behavioral margins—such as attention, memory, emotional activation, or social transmission—it generates larger political returns. In this regard, rigorous behavioral work, especially using carefully designed survey and experimental methods, could be particularly informative. On the other hand, further research could investigate the welfare implications of causal rhetoric. Fine-grained individual-level data could help quantify the distortions that causal rhetoric may impose on citizens' beliefs, and assess the extent to which its growing prominence contributes to the broader polarization of recent years.

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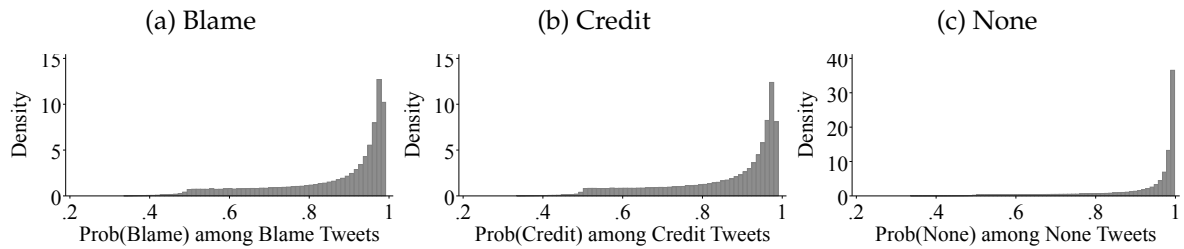
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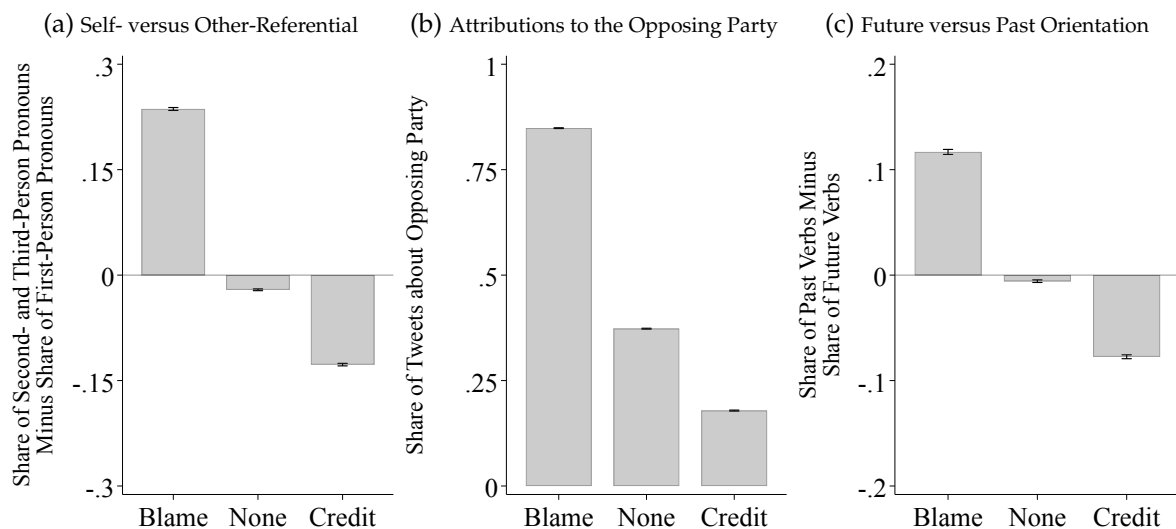
A Additional Figures and Tables

Figure A1: Distributions



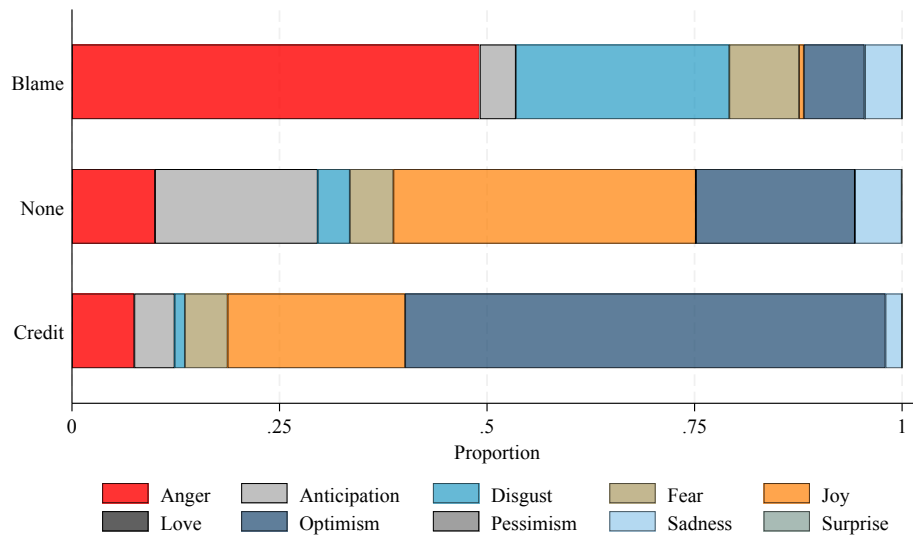
Notes: Each panel reports the distribution of the classifier-assigned probability for the predicted class, conditional on that class being selected. Panel (a) shows the distribution of $P(\text{Blame})$ among tweets classified as blame, that is, tweets for which $P(\text{Blame}) > \max\{P(\text{Credit}), P(\text{None})\}$. Panels (b) and (c) report the analogous distributions for credit and none.

Figure A2: Linguistic Features



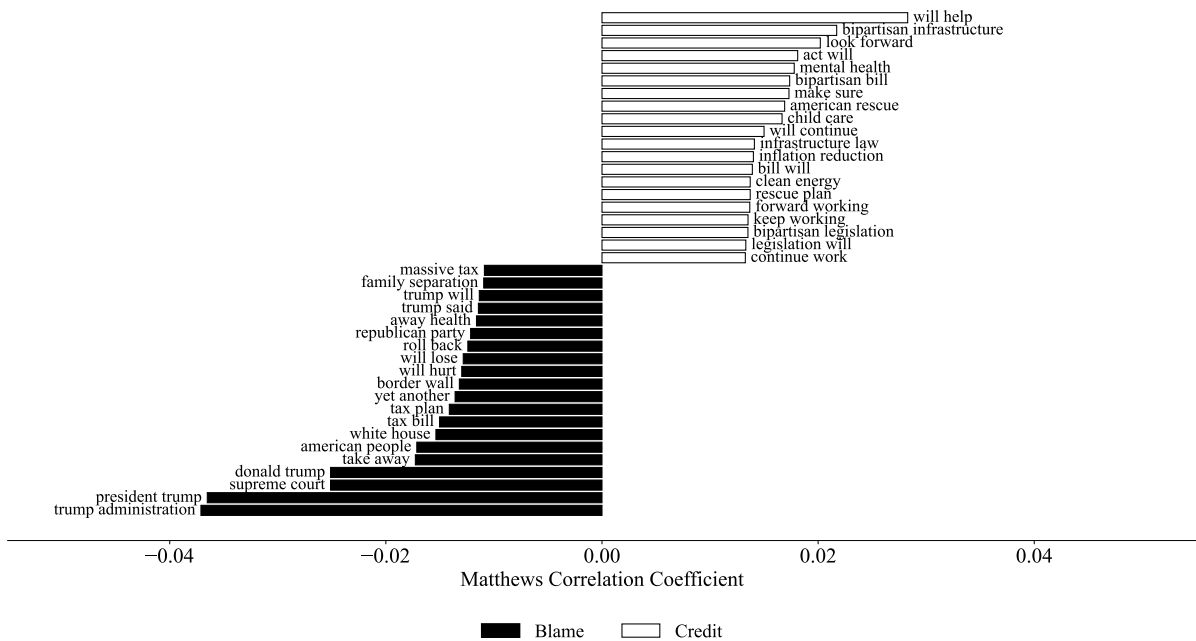
Notes: The figure reports linguistic and semantic measures across rhetorical styles. Panel (a) compares the relative prevalence of second- and third-person pronouns to first-person pronouns, capturing whether tweets are more other- or self-referential. Panel (b) reports the share of tweets that targets the opposing party, among tweets for which party identity can be identified. Panel (c) compares the use of future and past tense, capturing whether the rhetoric is more prospective or retrospective. The measures in panels (a) and (c) are standardized. Bars represent 95% confidence intervals.

Figure A3: Emotions



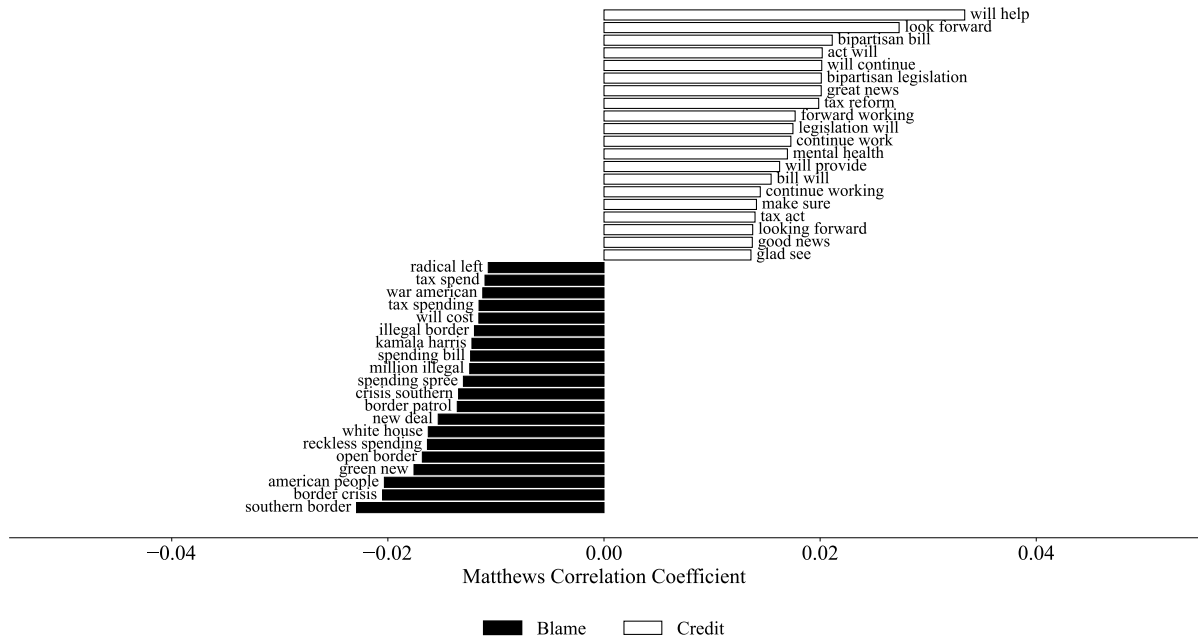
Notes: Each bar reports the share of tweets within a given rhetorical style classified into the corresponding emotion category.

Figure A4: Bigrams Diagnostic of Blame and Credit for Democrats



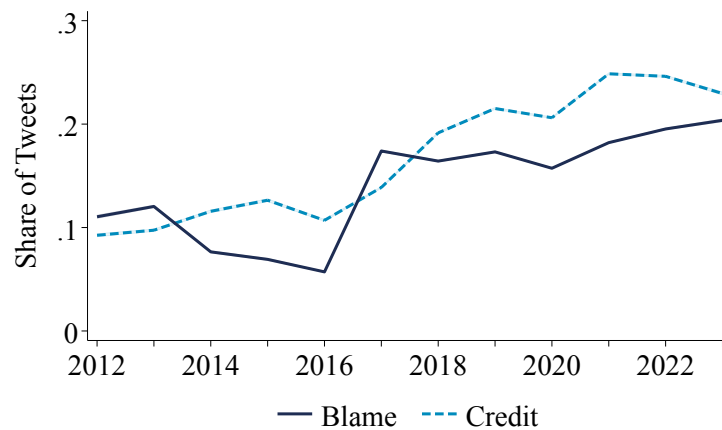
Notes: The figure reports the 20 bigrams that are most distinctive of blame and credit tweets among Democratic politicians, ranked by their Matthews Correlation Coefficient.

Figure A5: Bigrams Diagnostic of Blame and Credit for Republicans



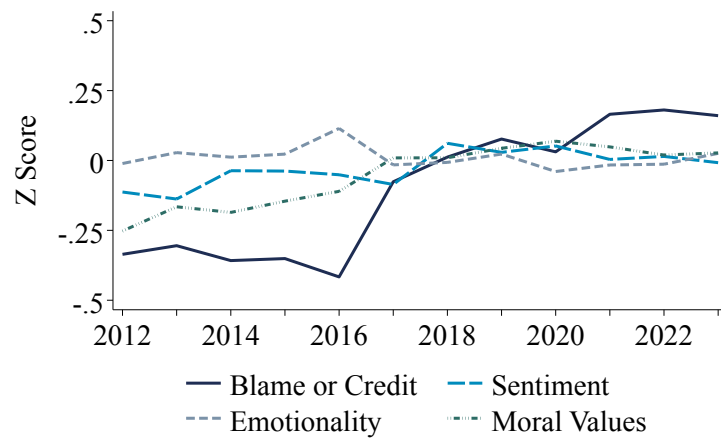
Notes: The figure reports the 20 bigrams that are most distinctive of blame and credit tweets among Republican politicians, ranked by their Matthews Correlation Coefficient.

Figure A6: Blame and Credit over Time, Excluding Senators



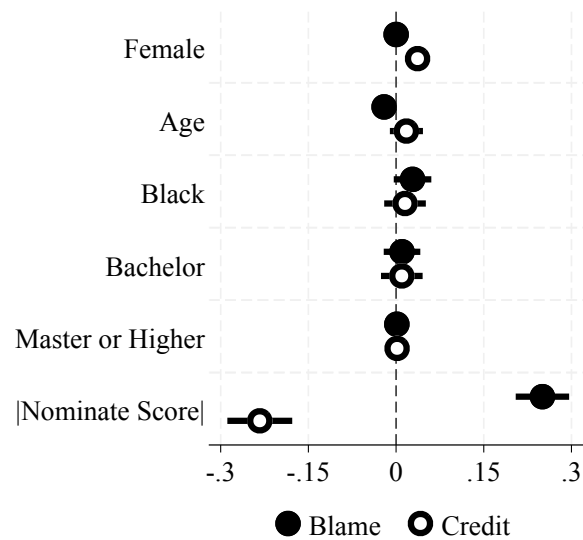
Notes: The figure presents the yearly share of tweets classified as blame and credit, excluding tweets from Senators. Shaded areas represent 95 percent confidence intervals.

Figure A7: Blame and Credit over Time, Comparison



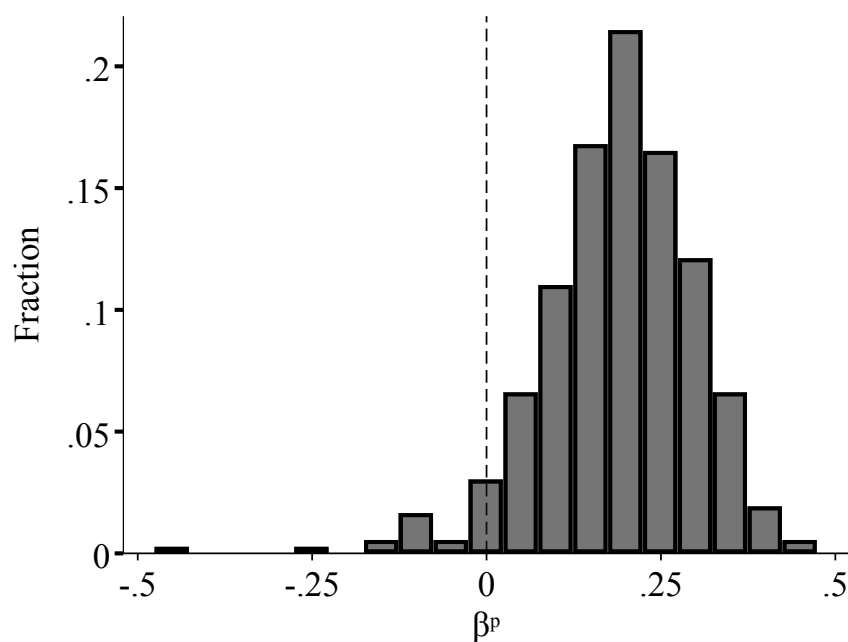
Notes: All text measures are standardized. Shaded areas represent 95 percent confidence intervals.

Figure A8: Author Correlates of Blame and Credit



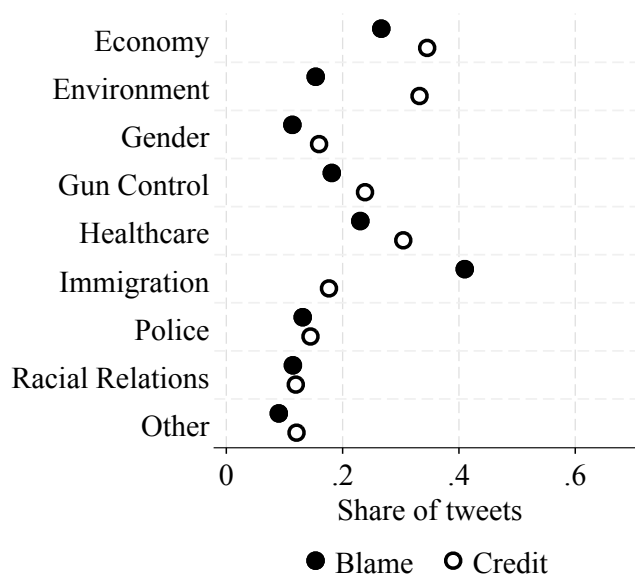
Notes: The coefficients presents the estimates from three regressions at the politician level of blame (credit) over the listed variables. Bars represent 95 percent confidence interval with robust standard errors.

Figure A9: Individual Increases in Causal Rhetoric



Notes: The coefficient reports the distribution of $\{\beta_i\}_i$ from the regression $y_{it} = \alpha + \beta_i D_{it}^{2017} + \varepsilon_{it}$ where y_{it} is binary indicator denoting whether tweet i is either blame or credit, D_{it}^{2017} is binary indicator taking the value 1 if tweet i is posted in or after 2017, estimated separately for each politician who appears in the dataset before and in or after 2017.

Figure A10: Blame and Credit over Topics



Notes: The figure presents the share of blame and credit tweets within each topic. Bars represent 95% confidence intervals.

Figure A11: Blame and Credit over Topics and Time

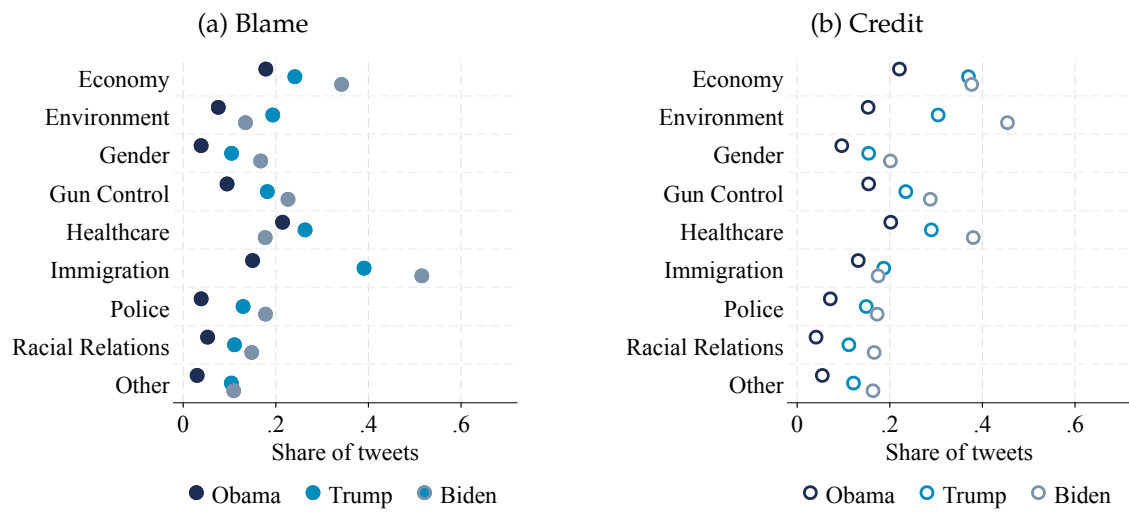


Figure A12: Elite Polarization Restricting to Targeted Tweets

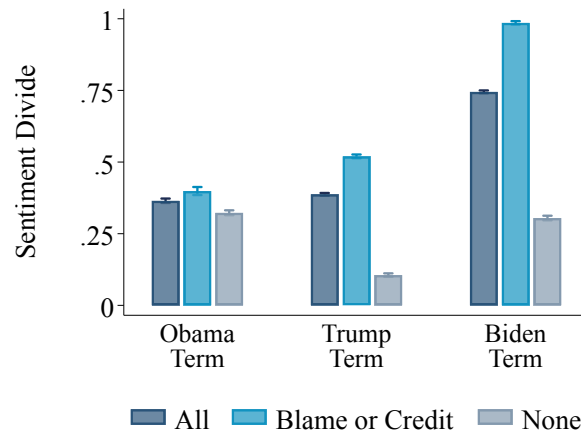


Table A1: Descriptive Statistics

	Panel a) Tweet Level			Panel b) Politician Level		
	All	Democrats	Republicans	All	Democrats	Republicans
Female	0.300 (0.458)	0.401 (0.490)	0.165 (0.371)	0.243 (0.429)	0.378 (0.486)	0.132 (0.339)
Age	57.973 (11.572)	59.058 (12.075)	56.509 (10.682)	57.468 (11.767)	58.499 (12.649)	56.617 (10.926)
Black	0.082 (0.274)	0.127 (0.333)	0.021 (0.142)	0.080 (0.271)	0.150 (0.357)	0.022 (0.148)
Bachelor	0.333 (0.471)	0.297 (0.457)	0.381 (0.486)	0.332 (0.471)	0.295 (0.457)	0.363 (0.481)
Master or Higher	0.620 (0.485)	0.672 (0.469)	0.551 (0.497)	0.584 (0.493)	0.651 (0.477)	0.529 (0.500)
Republican	0.426 (0.494)	0 (0)	1 (0)	0.548 (0.498)	0 (0)	1 (0)
Nominate Score	-0.005 (0.467)	-0.387 (0.126)	0.511 (0.164)	0.109 (0.455)	-0.366 (0.122)	0.500 (0.162)
Representative	0.804 (0.397)	0.814 (0.389)	0.790 (0.407)	0.857 (0.351)	0.857 (0.350)	0.856 (0.351)
Senator				0.118 (0.323)	0.123 (0.329)	0.114 (0.318)
Both Representative and Senator				0.026 (0.158)	0.020 (0.139)	0.030 (0.172)
Number of Tweets Posted				4664.950 (4848.342)	5924.391 (5151.747)	3625.209 (4319.150)
Number of Accounts				1.988 (0.689)	2.042 (0.688)	1.943 (0.687)
Observations	4,198,455	2,411,227	1,787,228	900	407	493

Notes: Panel a) presents statistics at the tweet level. Panel b) presents statistics at the politician level. Means and standard deviations in parentheses.

Table A2: Labeled Tweets

Tweet	Causal	Tone	Rhetoric Style
Biden has lost all credibility. No one believes your lies, Joe! URL	0	-1	None
.realDonaldTrump is a terrible role model for our kids. They deserve better. #Debate	0	-1	None
Joe Biden and the Democrats terrible policies have wreaked havoc on this country.	1	-1	Blame
#Trumpcare is fundamentally flawed. Higher costs, less coverage, fewer protections – thats GOPs gift to American people. #ProtectOurCare URL	1	-1	Blame
I couldnt support final passage of todays approps package but Im pleased about the inclusion of my HBCU amendment URL	0	1	None
Universal congrats to the scientists at @OregonState for their work helping Insight make a successful #MarsLanding: URL	0	1	None
The Protect Medical Innovation Act will boost American innovation and manufacturing, and it will encourage medical research and development that make a real difference in peoples lives. URL	1	1	Credit
Now that the Inflation Reduction Act is law it will not only lower prescription drug prices but save lives. Thank you @HenryFordHealth for your support. URL	1	1	Credit

Table A3: Democrats Tweets Classified as Blame and Credit

Panel a) Blame Tweets We warned when the GOP passed tax cuts for the rich that it would explode deficits. It did. We warned that the GOP would use those deficits to come after Social Security and Medicare. They are. URL The party of NO, #ILGOP in particular, plays political games in ignoring the implications on our economy, on jobs, Social Security checks. Republicans raised the debt ceiling 3 times under Trump's thumb. They are playing politics with people's lives. URL GOPs reckless health care strategy is already destabilizing #healthcare markets and forcing premiums to rise. If #GOP could govern, we wouldn't be having this conversation. The root of #TrumpShutdown is incompetence, putting veterans and kids health care at risk, pensions at risk, and the fight against opioids at risk Real wages today are lower than they were in 1973. Thats not a sign of a healthy economy, its a sign that working people today are worse off than they were 45 years ago, and the GOP tax cuts have done nothing to address that issue.
Panel b) Credit Tweets Proud to cosponsor @NydiaVelazquez's Public Housing Emergency Response Act which would invest \$70B in public housing including \$32B for @NYCHA. Our public housing crisis must be addressed & this bill is a bold approach to doing that. Residents deserve to live in safe conditions! I applaud @POTUS for setting our next offshore wind target. With a new infusion of investments from my offshore wind manufacturing tax credit in the Inflation Reduction Act, the U.S. can and will deploy 15 GW of floating offshore wind by 2035 all while creating good union jobs. URL QT @ginamccarthy46 Today we're announcing actions to advance *floating* offshore wind platforms – key to harnessing the potential of deep waters along the West Coast, Gulf of Maine, & more. Part of @POTUS' plan for American jobs and leadership on new clean energy technologies! URL The Bipartisan Infrastructure Law is putting our economy on track to thrive & investing in communities that have too often been left behind. With over \$2 million recently announced, we take a major step toward redeveloping Baltimores Bridge to Nowhere. URL Happy to be joining @HouseDemocrats to help Americas workers access better paying jobs. The Workforce Innovation and Opportunity Act connects employers with qualified candidates, lowers costs for families and increases supplies. Democrats are #InvestingInWorkers. URL The #BuildBackBetterAct provides much needed funds to @TheJusticeDept to help reduce community violence & fund proven intervention programs. Im proud to advocate for legislation to break cycles of violence in communities, saving American lives & taxpayer dollars.

Table A4: Republicans Tweets Classified as Blame and Credit

Panel a) Blame Tweets The supply chain and inflation crises are not a high class problem like @WHCOS claims. As Dems look to pour trillions into the economy and spike inflation further, they must understand that actions have consequences that will be felt by every American URL Our country is facing soaring inflation thanks to Democrats spending spree, and whats @POTUS solution? Spend MORE money. @JoeBiden and @SenateDems are TOTALLY out of touch with reality. Inflation is still wiping out wage growth, all while Democrats' reckless spending spree makes matters worse. URL Top border officials told Biden that if he unraveled Trump's policies and pushed for open borders that a major crisis would occur. He didnt listen. Now everyone is suffering – Americans and migrants alike. URL April saw the highest number of migrants ever recorded. Next week, @JoeBiden will reverse another commonsense border policy that will only make this crisis worse. Biden needs to wake up and face reality. URL
Panel b) Credit Tweets I'm an original sponsor of the Nuclear Energy Leadership Act w/ @RepElaineLuria to encourage further dev of advanced nuclear energy programs. Such programs will create high-quality jobs, strengthen natl security, reduce foreign energy dependence and promote emissions-free energy. Today @realDonaldTrump showed his commitment to supporting American energy dominance. The @EPA's rule will bolster our nations energy independence by lowering energy costs, spurring job growth and promoting economic development in our communities. URL Glad to hear @realDonaldTrump has signed legislation adding \$320 billion to the #PaycheckProtectionProgram, ramping up testing capability and providing more funding for health care providers. AR will benefit from this measure to protect public health and save businesses & jobs. URL URL Earlier this week, I introduced #CARA2 to increase funding levels for programs we know work and implements additional policy reforms that will make a real difference in combatting the #opioidcrisis. URL @TransportGOP are delivering on our promise of fixing supply chain holes and building a stronger economy. Currently we are marking up a package of bills that will remove barriers, increase efficiency, and target infrastructure investment. #SupplyChain

Table A5: List of Underlying County-Level Covariates

Variable
Population density
Log(County area)
Distance from Austin, TX (in miles)
Distance from NYC (in miles)
Distance from San Francisco (in miles)
Distance from Washington, DC (in miles)
% aged 20–24
% aged 25–29
% aged 30–34
% aged 35–39
% aged 40–44
% aged 45–49
% aged 50+
Population growth, 2000–2016
% white
% black
% native American
% Asian
% Hispanic
% unemployed
% below poverty level
% employed in IT
% employed in construction/real estate
% employed in manufacturing
% with high school degree
% with college education
% watching Fox News
% watching prime time TV

Notes: The table lists the 28 county-level covariates used to construct the principal components that are interacted with blame, credit, and tweet volume in Equations 3.

Table A6: Donations

	(1)	(2)	(3)	(4)
Panel a) Revenue from Donations				
Blame x Users	0.007*** (0.001)	0.006*** (0.001)	0.006*** (0.001)	0.007*** (0.002)
Credit x Users	0.001* (0.000)	0.001** (0.000)	0.002*** (0.000)	0.002* (0.001)
Blame x SXSWFollower2006	-0.008 (0.005)	-0.007* (0.004)	-0.007 (0.005)	-0.003 (0.003)
Credit x SXSWFollower2006	0.003* (0.002)	0.003 (0.002)	0.003 (0.002)	0.003* (0.002)
Panel b) Number of Donors				
Blame x Users	0.004*** (0.001)	0.003*** (0.000)	0.003*** (0.001)	0.004*** (0.001)
Credit x Users	-0.000 (0.000)	0.000 (0.000)	0.000*** (0.000)	0.000 (0.000)
Blame x SXSWFollower2006	-0.003 (0.002)	-0.002 (0.002)	-0.003 (0.002)	0.000 (0.002)
Credit x SXSWFollower2006	0.001 (0.001)	0.001 (0.001)	0.001 (0.001)	0.000 (0.001)
Panel c) Average per Donor				
Blame x Users	0.004*** (0.001)	0.003*** (0.001)	0.003*** (0.001)	0.003* (0.002)
Credit x Users	0.001* (0.000)	0.001** (0.000)	0.001*** (0.000)	0.002** (0.001)
Blame x SXSWFollower2006	-0.005 (0.003)	-0.005* (0.003)	-0.004 (0.003)	-0.004 (0.002)
Credit x SXSWFollower2006	0.002* (0.001)	0.002 (0.001)	0.003* (0.001)	0.003** (0.001)
Politician x Month FE	✓	✓	✓	✓
Politician x County FE	✓	✓	✓	✓
County x Month FE	✓	✓	✓	✓
Tweet Volume Control		✓	✓	✓
Sentiment Control			✓	
County Covariates Controls				✓
Observations	168,232,932	168,232,932	168,232,932	168,178,803
Clusters	3,108	3,108	3,108	3,107
F statistic	95.47	63.65	38.19	15.74

Notes: The table presents the 2SLS estimates of Equation 3. In Panel a) the outcome is the log-donation revenue. In Panel b) the outcome is the log-number of donors. In Panel c) the outcome is the log-average donation per donor. In each panel, the outcome is regressed on the monthly share of blame and credit tweets posted by the politician interacted with the log-number of Twitter users in the county. We also control for the same tweet shares interacted with the log-number of SXSW followers in the county who joined Twitter in 2006. The interactions with Twitter users in the county are instrumented using the corresponding interactions with SXSW followers in the county who joined Twitter in 2007. Standard errors in parentheses, clustered at the county level. *, **, *** denote significance at the 10%, 5%, and 1% levels, respectively.

Table A7: Donations OLS

	(1)	(2)	(3)	(4)
Panel a) Revenue from Donations				
Blame x Users	0.008*** (0.000)	0.006*** (0.000)	0.007*** (0.000)	0.007*** (0.000)
Credit x Users	-0.001*** (0.000)	-0.000*** (0.000)	0.000*** (0.000)	-0.001*** (0.000)
Panel b) Number of Donors				
Blame x Users	0.003*** (0.000)	0.003*** (0.000)	0.003*** (0.000)	0.003*** (0.000)
Credit x Users	-0.000*** (0.000)	-0.000*** (0.000)	0.000*** (0.000)	-0.001*** (0.000)
Panel c) Average per Donor				
Blame x Users	0.005*** (0.000)	0.004*** (0.000)	0.005*** (0.000)	0.005*** (0.000)
Credit x Users	-0.001*** (0.000)	-0.000*** (0.000)	0.000 (0.000)	-0.001*** (0.000)
Politician x Month FE	✓	✓	✓	✓
Politician x County FE	✓	✓	✓	✓
County x Month FE	✓	✓	✓	✓
Tweet Volume Control		✓	✓	✓
Sentiment Control			✓	
County Covariates Controls				✓
Observations	168,232,932	168,232,932	168,232,932	168,178,803
Clusters	3,108	3,108	3,108	3,107

Notes: The table presents the OLS estimates of Equation 3. In Panel a) the outcome is the log-donation revenue. In Panel b) the outcome is the log-number of donors. In Panel c) the outcome is the log-average donation per donor. In each panel, the outcome is regressed on the monthly share of blame and credit tweets posted by the politician interacted with the log-number of Twitter users in the county. Standard errors in parentheses, clustered at the county level. *, **, *** denote significance at the 10%, 5%, and 1% levels, respectively.

Table A8: Donations Reduced Form

	(1)	(2)	(3)	(4)
Panel a) Revenue from Donations				
Blame x SXSWFollower2007	0.018*** (0.003)	0.014*** (0.003)	0.016*** (0.003)	0.007** (0.003)
Credit x SXSWFollower2007	0.002* (0.001)	0.003** (0.001)	0.004*** (0.001)	0.002** (0.001)
Blame x SXSWFollower2006	-0.010 (0.007)	-0.009 (0.005)	-0.009 (0.006)	-0.010** (0.005)
Credit x SXSWFollower2006	0.003 (0.002)	0.003 (0.002)	0.003 (0.002)	0.001 (0.002)
Panel b) Number of Donors				
Blame x SXSWFollower2007	0.009*** (0.002)	0.007*** (0.001)	0.008*** (0.002)	0.005*** (0.001)
Credit x SXSWFollower2007	-0.000 (0.000)	0.000 (0.000)	0.001*** (0.000)	0.000 (0.000)
Blame x SXSWFollower2006	-0.004 (0.003)	-0.003 (0.002)	-0.004 (0.003)	-0.004* (0.002)
Credit x SXSWFollower2006	0.001 (0.001)	0.001 (0.001)	0.001 (0.001)	0.000 (0.001)
Panel c) Average per Donor				
Blame x SXSWFollower2007	0.010*** (0.002)	0.008*** (0.002)	0.008*** (0.002)	0.003* (0.002)
Credit x SXSWFollower2007	0.002* (0.001)	0.002** (0.001)	0.002*** (0.001)	0.002** (0.001)
Blame x SXSWFollower2006	-0.007 (0.004)	-0.006 (0.003)	-0.005 (0.004)	-0.006** (0.003)
Credit x SXSWFollower2006	0.002 (0.001)	0.002 (0.001)	0.002 (0.002)	0.001 (0.001)
Politician x Month FE	✓	✓	✓	✓
Politician x County FE	✓	✓	✓	✓
County x Month FE	✓	✓	✓	✓
Tweet Volume Control		✓	✓	✓
Sentiment Control			✓	
County Covariates Controls				✓
Observations	168,232,932	168,232,932	168,232,932	168,178,803
Clusters	3,108	3,108	3,108	3,107

Notes: The table presents the reduced form estimates of Equation 3. In Panel a) the outcome is the log-donation revenue. In Panel b) the outcome is the log-number of donors. In Panel c) the outcome is the log-average donation per donor. In each panel, the outcome is regressed on the monthly share of blame and credit tweets posted by the politician interacted with the log-number of SXSW followers in the county who joined Twitter in 2007. We also control for the same tweet shares interacted with the log-number of SXSW followers in the county who joined Twitter in 2006. Standard errors in parentheses, clustered at the county level.

*, **, *** denote significance at the 10%, 5%, and 1% levels, respectively.

Table A9: Donations First Stage

	(1)	(2)	(3)	(4)
Panel a) Blame x Users				
Blame x SXSWFollower2007	2.482*** (0.180)	2.482*** (0.180)	2.482*** (0.180)	1.068*** (0.155)
Credit x SXSWFollower2007	-0.000*** (0.000)	-0.000*** (0.000)	-0.000 (0.000)	0.000*** (0.000)
Blame x SXSWFollower2006	-0.274 (0.394)	-0.274 (0.394)	-0.274 (0.394)	-0.912*** (0.337)
Credit x SXSWFollower2006	0.000 (0.000)	0.000 (0.000)	0.000 (0.000)	0.000** (0.000)
Panel b) Credit x Users				
Blame x SXSWFollower2007	-0.000*** (0.000)	-0.000*** (0.000)	-0.000*** (0.000)	0.000 (0.000)
Credit x SXSWFollower2007	2.482*** (0.180)	2.482*** (0.180)	2.482*** (0.180)	1.068*** (0.155)
Blame x SXSWFollower2006	0.000 (0.000)	0.000 (0.000)	0.000 (0.000)	0.000** (0.000)
Credit x SXSWFollower2006	-0.274 (0.394)	-0.274 (0.394)	-0.274 (0.394)	-0.912*** (0.337)
Politician x Month FE	✓	✓	✓	✓
Politician x County FE	✓	✓	✓	✓
County x Month FE	✓	✓	✓	✓
Tweet Volume Control		✓	✓	✓
Sentiment Control			✓	
County Covariates Controls				✓
Observations	168,232,932	168,232,932	168,232,932	168,178,803
Clusters	3,108	3,108	3,108	3,107

Notes: The table presents the first-stage regressions related to the 2SLS estimation of Equation 3. In Panel a) the outcome is the monthly share of blame tweets posted by the politician interacted with the log-number of Twitter users in the county. In Panel b) the outcome is the monthly share of credit tweets posted by the politician interacted with the log-number of Twitter users in the county. In each panel, the outcome is regressed on the monthly share of blame and credit tweets posted by the politician interacted with the log-number of SXSW followers in the county who joined Twitter in 2007. The table also includes the estimates for the same tweet shares interacted with the log-number of SXSW followers in the county who joined Twitter in 2006. Standard errors in parentheses, clustered at the county level. *, **, *** denote significance at the 10%, 5%, and 1% levels, respectively.

Table A10: Probability of Receiving a Donation

	(1)	(2)	(3)	(4)
Blame x Users	0.002*** (0.000)	0.002*** (0.000)	0.001*** (0.000)	0.002*** (0.001)
Credit x Users	-0.000* (0.000)	-0.000 (0.000)	-0.000 (0.000)	-0.000 (0.000)
Blame x SXSWFollower2006	-0.001 (0.001)	-0.001 (0.001)	-0.001 (0.001)	-0.000 (0.001)
Credit x SXSWFollower2006	0.001 (0.000)	0.000 (0.000)	0.000 (0.000)	0.000 (0.000)
Politician x Month FE	✓	✓	✓	✓
Politician x County FE	✓	✓	✓	✓
County x Month FE	✓	✓	✓	✓
Tweet Volume Control		✓	✓	✓
Sentiment Control			✓	
County Covariates Controls				✓
Observations	168,232,932	168,232,932	168,232,932	168,178,803
Clusters	3,108	3,108	3,108	3,107
F statistic	95.47	63.65	38.19	15.74

Notes: The table presents the 2SLS estimates of Equation 3. The outcome is a binary indicator equal to 1 if the politician receives at least one donation from a given county in a given month, and 0 otherwise. The outcome is regressed on the monthly shares of blame and credit tweets posted by the politician, each interacted with the log-number of Twitter users in the county. We also control for the corresponding interactions with the log-number of SXSW followers in the county who joined Twitter in 2006. The interactions with Twitter users in the county are instrumented using the corresponding interactions with the log-number of SXSW followers in the county who joined Twitter in 2007. Standard errors in parentheses, clustered at the county level. *, **, *** denote significance at the 10%, 5%, and 1% levels, respectively.

Table A11: Donations from Ideologically Moderate Donors

	(1)	(2)	(3)	(4)
Panel a) Revenue from Donations				
Blame x Users	-0.002*** (0.000)	-0.003*** (0.001)	-0.003*** (0.001)	-0.003* (0.001)
Credit x Users	0.004*** (0.001)	0.005*** (0.001)	0.004*** (0.001)	0.006*** (0.002)
Blame x SXSWFollower2006	-0.001 (0.002)	-0.001 (0.003)	-0.000 (0.003)	0.000 (0.002)
Credit x SXSWFollower2006	-0.000 (0.003)	-0.000 (0.003)	0.001 (0.003)	0.001 (0.003)
Panel b) Number of Donors				
Blame x Users	-0.001*** (0.000)	-0.002*** (0.000)	-0.002*** (0.000)	-0.002*** (0.001)
Credit x Users	0.002*** (0.000)	0.002*** (0.000)	0.002*** (0.000)	0.003*** (0.001)
Blame x SXSWFollower2006	-0.000 (0.001)	-0.000 (0.001)	-0.000 (0.001)	-0.000 (0.001)
Credit x SXSWFollower2006	-0.000 (0.001)	-0.000 (0.001)	-0.000 (0.001)	0.001 (0.001)
Panel c) Average per Donor				
Blame x Users	-0.000 (0.000)	-0.001*** (0.000)	-0.001*** (0.000)	-0.000 (0.001)
Credit x Users	0.002*** (0.000)	0.003*** (0.000)	0.002*** (0.000)	0.003*** (0.001)
Blame x SXSWFollower2006	-0.001 (0.001)	-0.000 (0.002)	0.000 (0.002)	0.001 (0.001)
Credit x SXSWFollower2006	0.000 (0.002)	0.000 (0.002)	0.001 (0.002)	0.001 (0.002)
Politician x Month FE	✓	✓	✓	✓
Politician x County FE	✓	✓	✓	✓
County x Month FE	✓	✓	✓	✓
Tweet Volume Control		✓	✓	✓
Sentiment Control			✓	
County Covariates Controls				✓
Observations	168,232,932	168,232,932	168,232,932	168,178,803
Clusters	3,108	3,108	3,108	3,107
F statistic	95.47	63.65	38.19	15.74

Notes: The table presents the 2SLS estimates of Equation 3 restricting to donations from ideologically moderate donors. In Panel a) the outcome is the log-donation revenue. In Panel b) the outcome is the log-number of donors. In Panel c) the outcome is the log-average donation per donor. In each panel, the outcome is regressed on the monthly share of blame and credit tweets posted by the politician interacted with the log-number of Twitter users in the county. We also control for the same tweet shares interacted with the log-number of SXSW followers in the county who joined Twitter in 2006. The interactions with Twitter users in the county are instrumented using the corresponding interactions with SXSW followers in the county who joined Twitter in 2007. Standard errors in parentheses, clustered at the county level. *, **, *** denote significance at the 10%, 5%, and 1% levels, respectively.

Table A12: Donations from Ideologically Extreme Donors

	(1)	(2)	(3)	(4)
Panel a) Revenue from Donations				
Blame x Users	0.012*** (0.002)	0.011*** (0.002)	0.012*** (0.002)	0.014*** (0.004)
Credit x Users	-0.004*** (0.001)	-0.004*** (0.001)	-0.002*** (0.000)	-0.004*** (0.001)
Blame x SXSWFollower2006	-0.006 (0.008)	-0.006 (0.007)	-0.006 (0.008)	-0.001 (0.006)
Credit x SXSWFollower2006	0.003 (0.002)	0.003 (0.002)	0.003* (0.002)	0.001 (0.002)
Panel b) Number of Donors				
Blame x Users	0.006*** (0.001)	0.005*** (0.001)	0.006*** (0.001)	0.007*** (0.002)
Credit x Users	-0.002*** (0.000)	-0.002*** (0.000)	-0.001*** (0.000)	-0.003*** (0.001)
Blame x SXSWFollower2006	-0.002 (0.003)	-0.002 (0.003)	-0.003 (0.003)	0.001 (0.003)
Credit x SXSWFollower2006	0.001 (0.001)	0.001 (0.001)	0.001 (0.001)	-0.000 (0.001)
Panel c) Average per Donor				
Blame x Users	0.008*** (0.001)	0.007*** (0.001)	0.007*** (0.001)	0.007*** (0.003)
Credit x Users	-0.002*** (0.000)	-0.002*** (0.000)	-0.001*** (0.000)	-0.002** (0.001)
Blame x SXSWFollower2006	-0.004 (0.005)	-0.004 (0.005)	-0.004 (0.005)	-0.002 (0.004)
Credit x SXSWFollower2006	0.002 (0.002)	0.002 (0.002)	0.002* (0.001)	0.002 (0.001)
Politician x Month FE	✓	✓	✓	✓
Politician x County FE	✓	✓	✓	✓
County x Month FE	✓	✓	✓	✓
Tweet Volume Control		✓	✓	✓
Sentiment Control			✓	
County Covariates Controls				✓
Observations	168,232,932	168,232,932	168,232,932	168,178,803
Clusters	3,108	3,108	3,108	3,107
F statistic	95.47	63.65	38.19	15.74

Notes: The table presents the 2SLS estimates of Equation 3 restricting to donations from ideologically extreme donors. In Panel a) the outcome is the log-donation revenue. In Panel b) the outcome is the log-number of donors. In Panel c) the outcome is the log-average donation per donor. In each panel, the outcome is regressed on the monthly share of blame and credit tweets posted by the politician interacted with the log-number of Twitter users in the county. We also control for the same tweet shares interacted with the log-number of SXSW followers in the county who joined Twitter in 2006. The interactions with Twitter users in the county are instrumented using the corresponding interactions with SXSW followers in the county who joined Twitter in 2007. Standard errors in parentheses, clustered at the county level. *, **, *** denote significance at the 10%, 5%, and 1% levels, respectively.

Table A13: Revenue from Donations over Time

	Revenue from Donations	Number of Donors	Average per Donor
(Blame x Users) x Obama	0.000 (0.000)	-0.000*** (0.000)	0.001*** (0.000)
(Blame x Users) x Trump	0.002* (0.001)	0.001** (0.000)	0.001** (0.001)
(Blame x Users) x Biden	0.014*** (0.002)	0.008*** (0.001)	0.007*** (0.001)
(Credit x Users) x Obama	0.003*** (0.001)	0.001*** (0.000)	0.003*** (0.000)
(Credit x Users) x Trump	0.004*** (0.001)	0.002*** (0.000)	0.002*** (0.000)
(Credit x Users) x Biden	-0.003*** (0.001)	-0.002*** (0.000)	-0.002** (0.001)
(Blame x SXSWFollower2006) x Obama	-0.000 (0.002)	0.000 (0.001)	-0.001 (0.002)
(Blame x SXSWFollower2006) x Trump	-0.004 (0.003)	-0.001 (0.001)	-0.002 (0.002)
(Blame x SXSWFollower2006) x Biden	-0.015* (0.009)	-0.005 (0.004)	-0.010* (0.005)
(Credit x SXSWFollower2006) x Obama	0.002 (0.003)	0.001 (0.001)	0.001 (0.002)
(Credit x SXSWFollower2006) x Trump	-0.000 (0.003)	-0.000 (0.001)	0.000 (0.002)
(Credit x SXSWFollower2006) x Biden	0.006 (0.004)	0.002 (0.002)	0.005 (0.003)
Politician x Month FE	✓	✓	✓
Politician x County FE	✓	✓	✓
County x Month FE	✓	✓	✓
Tweet Volume Control	✓	✓	✓
Observations	168,232,932	168,232,932	168,232,932
Clusters	3,108	3,108	3,108
F statistic	21.22	21.22	21.22

Notes: The table presents the 2SLS estimates of Equation 3 allowing the coefficients of the terms involving the share of blame and the share of credit tweets to vary across presidencies. The outcomes are the log-donation revenue, the log-number of donors, and the log-average donation per donor in columns 1, 2, and 3, respectively. The outcome is regressed on the monthly share of blame and credit tweets posted by the politician interacted with the log-number of Twitter users in the county. We also control for the same tweet shares interacted with the log-number of SXSW followers in the county who joined Twitter in 2006. To allow for heterogeneity across presidencies, all interactions are further interacted with presidency indicators. The interactions with Twitter users in the county are instrumented using the corresponding interactions with SXSW followers in the county who joined Twitter in 2007. Standard errors in parentheses, clustered at the county level. *, **, *** denote significance at the 10%, 5%, and 1% levels, respectively.

Table A14: Blame, Credit, and Legislative Activity

	Tweet is Blame	Tweet is Credit
Bill Presented by Opposing Party	0.051*** (0.004)	0.002 (0.003)
Bill Presented by Own Party	-0.009*** (0.003)	0.058*** (0.003)
Politician FE	✓	✓
Week FE	✓	✓
Observations	4,198,452	4,198,452
Clusters	4,202	4,202

Notes: The table presents the estimates of Equation 4. In column (1), the outcome is a binary indicator equal to 1 if the tweet is classified as blame and 0 otherwise. In column (2), the outcome is a binary indicator equal to 1 if the tweet is classified as credit and 0 otherwise. In each column, the outcome is regressed on two binary indicators: one equal to 1 if a bill is introduced on that day by a member of the opposing party, and one equal to 1 if a bill is introduced on that day by a member of the politician's own party. All specifications include politician and week fixed effects. Standard errors in parentheses are clustered at the day level. *, **, *** denote significance at the 10%, 5%, and 1% levels, respectively.

Table A15: Blame, Credit, and Legislative Activity, Restricting within Related Sentiment

	Tweet is Blame	Tweet is Credit
Bill Presented by Opposing Party	0.051*** (0.005)	0.036*** (0.004)
Bill Presented by Own Party	-0.008* (0.005)	0.086*** (0.004)
Politician FE	✓	✓
Week FE	✓	✓
Observations	996,003	1,864,364
Clusters	4,202	4,202

Notes: The table presents the estimates of Equation 4. In column (1), the outcome is a binary indicator equal to 1 if the tweet is classified as blame and 0 otherwise; the sample is restricted to tweets classified as having negative sentiment. In column (2), the outcome is a binary indicator equal to 1 if the tweet is classified as credit and 0 otherwise; the sample is restricted to tweets classified as having positive sentiment. In each column, the outcome is regressed on two binary indicators: one equal to 1 if a bill is introduced on that day by a member of the opposing party, and one equal to 1 if a bill is introduced on that day by a member of the politician's own party. All specifications include politician and week fixed effects. Standard errors in parentheses are clustered at the day level. *, **, *** denote significance at the 10%, 5%, and 1% levels, respectively.

B Additional Results

B.1 Platform Changes

As discussed in Section 4, the supply of blame and credit tweets rises sharply over our sample period, especially after 2017. A natural concern is that this pattern may reflect platform-level changes implemented by Twitter rather than a genuine change in politicians' rhetorical choices. Two changes are particularly relevant. On February 10, 2016, Twitter introduced algorithmic feed curation, replacing the strictly chronological ordering of tweets. On November 7, 2017, Twitter expanded the character limit from 140 to 280 characters. We find no evidence that either change generated a discontinuous increase in the production of blame or credit tweets, making it unlikely that platform design is driving the patterns we document.

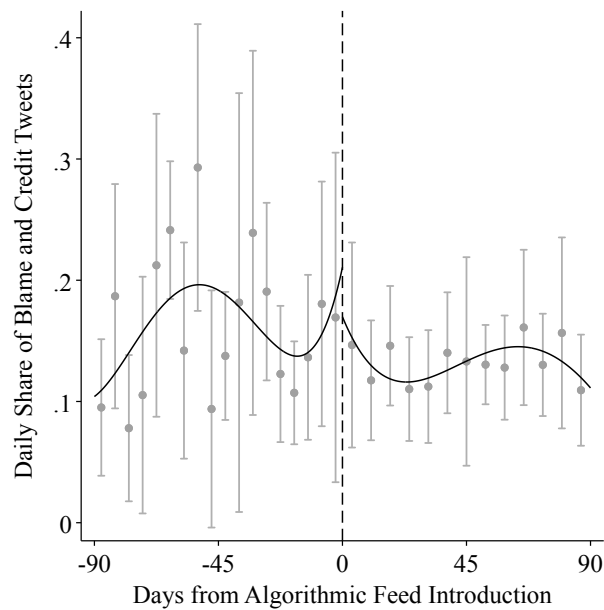
We first examine the introduction of the algorithmic feed. At a descriptive level, Figure 3 already suggests that the use of blame and credit does not increase noticeably in 2016. We then provide a more formal test using a regression discontinuity design centered on February 10, 2016, with the daily share of tweets classified as blame or credit as the outcome and a symmetric 90-day window around the policy change. As shown in Figure B1, there is no evidence of a discontinuity at the cutoff. The estimated jump is -0.035 and statistically insignificant ($p = 0.529$). This result suggests that the introduction of the algorithmic feed did not contribute meaningfully to the subsequent rise in causal rhetoric.

We next turn to the expansion of the character limit. A first question is whether the original 140-character constraint disproportionately restricted blame or credit tweets. To assess this, Figure B2 plots the distribution of tweet length, measured in characters, before November 7, 2017. The figure shows visible bunching near the upper bound, especially between 100 and 120 characters, indicating that many tweets were close to exhausting the character limit. Crucially, however, this pattern looks very similar across rhetorical categories: blame, credit, and none tweets all display comparable length distributions. If blame or credit had been especially constrained by the original cap, one would expect stronger bunching near the limit for those categories. Instead, their distributions closely resemble that of none tweets. This makes it unlikely that the increase to 280 characters mechanically enabled the later rise in blame or credit.

We then test this more formally using a regression discontinuity design around November 7, 2017, again using the daily share of tweets classified as blame or credit as the outcome and a symmetric 90-day window around the reform. As shown in Figure B3, there is no evidence of a discontinuity at the threshold. The estimated jump is 0.056 and statistically insignificant ($p = 0.532$). This reinforces the conclusion that

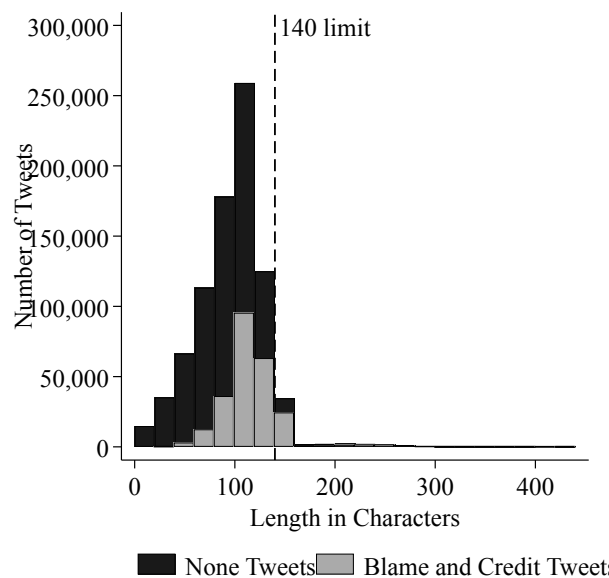
the character-limit expansion was not a major driver of the growth in causal rhetoric.

Figure B1: Regression Discontinuity Plot around Algorithmic Feed Introduction



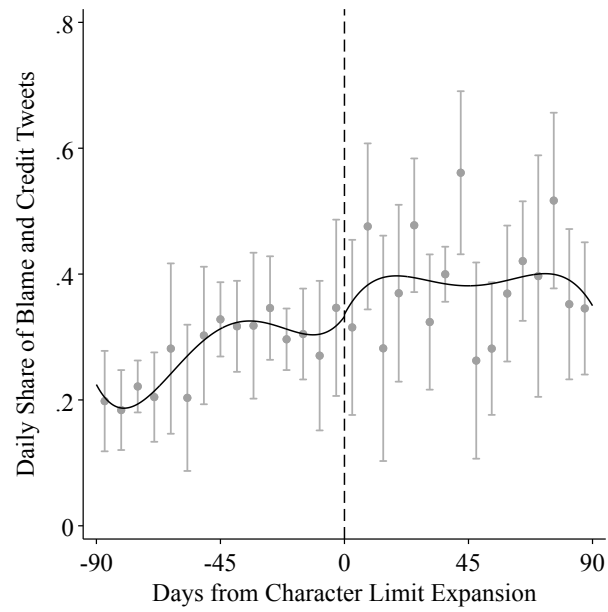
Notes: The figure reports the regression discontinuity plot around February 10, 2016, the date Twitter introduced algorithmic feed curation. The outcome is the daily share of tweets classified as blame or credit. The plot is produced using the `rdrobust` package of Calonico et al. (2017).

Figure B2: Distribution of Tweets Length before Character Limit Expansion



Notes: The figure shows the distribution of tweet length, measured in characters, for tweets posted before November 7, 2017, when Twitter expanded the character limit from 140 to 280.

Figure B3: Regression Discontinuity Plot around Character Limit Expansion



Notes: The figure reports the regression discontinuity plot around November 7, 2017, the date Twitter expanded the character limit from 140 to 280. The outcome is the daily share of tweets classified as blame or credit. The plot is produced using the `rdrobust` package of [Calonico et al. \(2017\)](#).

B.2 Protest Activity

This appendix provides additional evidence consistent with the main mobilization result in Section 5. There, we show that blame is uniquely effective at expanding support on the extensive margin by increasing the number of donors. Here, we ask whether a similar pattern appears in a broader form of political participation: protest activity.

We study this relationship using data from the [Crowd Counting Consortium](#), hosted by the Ash Center for Democratic Governance and Innovation at Harvard University. The dataset records protest events across the United States on a wide range of political and social causes. We use data from the first two phases of the project, covering 2017–2020 and 2021–2024. In total, the full dataset contains 212,004 protest events. In the analysis, we restrict attention to the period that overlaps with our Twitter data, so the effective sample runs through 2023.

To estimate the relationship between causal rhetoric and protest activity, we adapt the baseline specification from Section 5 to the county-month level:

$$y_{cm} = \beta_1(\text{Blame}_m \times \text{Users}_c) + \beta_2(\text{Credit}_m \times \text{Users}_c) + \gamma X_{cm} + \lambda_c + \mu_m + \varepsilon_{cm}. \quad (5)$$

Here, y_{cm} is an indicator equal to one if at least one protest occurs in county c and month m , and zero otherwise. Blame_m and Credit_m denote the aggregate monthly shares of blame and credit tweets posted by politicians nationwide, while Users_c measures Twitter penetration in county c . As in Section 5, our preferred specification includes only tweet volume interacted with county Twitter penetration, while richer specifications additionally control for non-causal sentiment and for interactions with principal components constructed from county-level covariates.

The identification logic follows the donation analysis. Counties with greater Twitter penetration are more exposed to the aggregate rhetorical environment on the platform. We therefore ask whether protest activity responds differentially to monthly variation in blame and credit in places where exposure to Twitter is higher. As before, we instrument county Twitter penetration with the number of followers of the official SXSW Twitter account in 2007, and instrument the interaction terms using the corresponding interactions with SXSW followers in 2007. We also include control interactions with pre-2007 SXSW followers, paralleling the design used in Section 5.

Table B1 reports the results and Table B2 reports the first-stage estimates. In our preferred specification, a one-standard-deviation increase in the monthly share of blame tweets posted by politicians raises the probability that a county with average Twitter

penetration experiences at least one protest by about 4.2 percentage points. Across specifications, the implied effect ranges from 4.2 to 10.6 percentage points. Credit, by contrast, has no robust positive effect.

These results corroborate the main evidence in the paper. In the donation data, blame is the only rhetorical style that robustly mobilizes participation on the extensive margin. The protest results point in the same direction: beyond increasing the number of donors, blame is also associated with greater offline collective action. Although protest activity is observed at an aggregate rather than politician-specific level, this pattern is consistent with the main results and suggests that blame may be especially effective at activating participation more broadly.

Table B1: Protests

	(1)	(2)	(3)	(4)
Blame x Users	0.008*** (0.002)	0.008*** (0.002)	0.017*** (0.004)	0.020*** (0.006)
Credit x Users	0.003 (0.003)	0.002 (0.003)	0.007* (0.004)	-0.009 (0.007)
Blame x SXSWFollower2006	-0.019** (0.009)	-0.020** (0.009)	-0.038** (0.015)	-0.002 (0.007)
Credit x SXSWFollower2006	0.003 (0.012)	0.009 (0.012)	-0.003 (0.016)	-0.010 (0.009)
County FE	✓	✓	✓	✓
Month FE	✓	✓	✓	✓
Tweet Volume Control		✓	✓	✓
Sentiment Control			✓	
County Covariates Controls				✓
Observations	245,532	245,532	245,532	245,453
Clusters	3,108	3,108	3,108	3,107
F statistic	95.47	63.65	38.19	15.74

Notes: The table presents the 2SLS estimates of Equation 5. The outcome is an indicator equal to one if at least one protest occurs in county *and* month, and zero otherwise. The outcome is regressed on the monthly shares of blame and credit tweets interacted with the log-number of Twitter users in the county. We also control for the same tweet shares interacted with the log-number of SXSW followers in the county who joined in 2006. The interactions with Twitter users are instrumented using the corresponding interaction with SXSW followers in the county who joined in 2007. Standard errors in parentheses clustered at the county level. *, **, *** denote significance at the 10%, 5%, and 1% levels, respectively.

Table B2: Protests First Stage

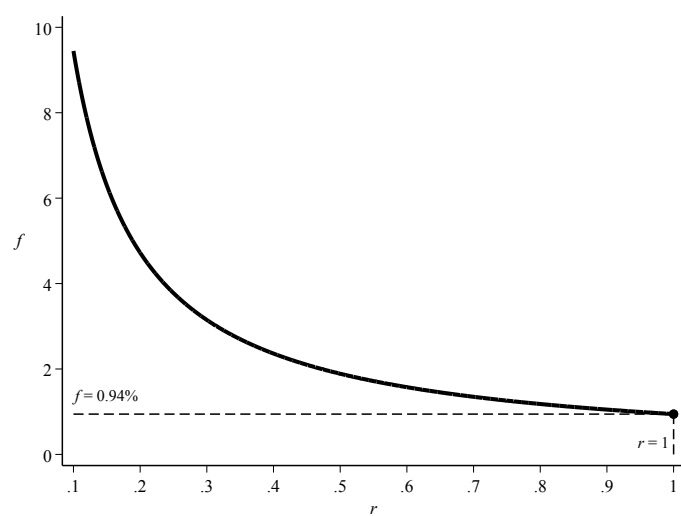
	(1)	(2)	(3)	(4)
Panel a) Blame x Users				
Blame x SXSWFollower2007	2.482*** (0.180)	2.482*** (0.180)	2.482*** (0.180)	1.068*** (0.155)
Credit x SXSWFollower2007	0.000 (0.000)	-0.000 (0.000)	-0.000 (0.000)	0.000 (0.000)
Blame x SXSWFollower2006	-0.274 (0.394)	-0.274 (0.394)	-0.274 (0.394)	-0.912*** (0.337)
Credit x SXSWFollower2006	-0.000*** (0.000)	-0.000*** (0.000)	-0.000 (0.000)	-0.000*** (0.000)
Panel b) Credit x Users				
Blame x SXSWFollower2007	0.000*** (0.000)	0.000*** (0.000)	-0.000 (0.000)	0.000 (0.000)
Credit x SXSWFollower2007	2.482*** (0.180)	2.482*** (0.180)	2.482*** (0.180)	1.068*** (0.155)
Blame x SXSWFollower2006	0.000 (0.000)	0.000 (0.000)	0.000 (0.000)	0.000 (0.000)
Credit x SXSWFollower2006	-0.274 (0.394)	-0.274 (0.394)	-0.274 (0.394)	-0.912*** (0.337)
County FE	✓	✓	✓	✓
Month FE	✓	✓	✓	✓
Tweet Volume Control		✓	✓	✓
Sentiment Control			✓	
County Covariates Controls				✓
Observations	245,532	245,532	245,532	245,453
Clusters	3,108	3,108	3,108	3,107

Notes: The table presents the first-stage regressions related to the 2SLS estimation of Equation 5. In Panel a) the outcome is the monthly share of blame tweets posted by all politicians interacted with the log-number of Twitter users in the county. In Panel b) the outcome is the monthly share of credit tweets posted by all politicians interacted with the log-number of Twitter users in the county. In each panel, the outcome is regressed on the monthly share of blame and credit tweets posted by the politician interacted with the log-number of SXSW followers in the county who joined in 2007. The table also includes the estimates for the same tweet shares interacted with the log-number of SXSW followers in the county who joined in 2006. Standard errors in parentheses clustered at the county level. *, **, *** denote significance at the 10%, 5%, and 1% levels, respectively.

B.3 Persuasion Rate

To interpret the magnitude of our estimates, we calculate persuasion rates following DellaVigna and Gentzkow (2010) as $f = \frac{y_c - y_t}{r \times e_t - e_c} \times \frac{1}{1 - y_0}$. In the expression $y_c - y_t$ is the difference in the share of population donating between treated and untreated, y_0 is the share that would donate without the treatment, $e_t - e_c$ is the difference in exposure between the treated and untreated, and r is a reach parameter. In our setting, we can write $y_t - y_c = \bar{\beta} \times y_c = \beta \times \overline{\text{Users}} \times y_c$, where β is the semi-elasticity of log-number of donors to a specific rhetorical style and $\overline{\text{Users}}$ is the average county log Twitter penetration—5.29 in our data. Since the estimated β for credit is close to 0 and statistically insignificant, we focus on blame—for which, in our preferred specification, we estimate $\beta = 0.003$. As standard in the literature, we assume $y_0 = y_c$, and we set $y_c = 0.16$ following Boken et al. (2023). The $r \times e_t - e_c$ term deserves more discussion. While e_c is mechanically equal to 0, e_t represents the share of the average county that is on Twitter (again, 0.32 in Boken et al. (2023)). For our purposes, however, the relevant exposure metric is the share of the average county that could have been exposed to the rhetorical style. For instance, since they focus on virality, Boken et al. (2023) estimate assumes that a viral tweet will be seen by everybody who is on Twitter, effectively setting $r = 1$. This is the same assumption that leads to the 0.94% persuasion rate reported in the main text, which is a lower bound. However, there are reasons to believe that r could be lower. For instance, Wojcieszak et al. (2022) find that 40% of American Twitter users follow at least one politician. Assuming that users see only the posts of accounts they follow, and they see all of them, this would lead to $r = 0.4$ and thus $f = 2.36\%$. Hence, to be as transparent as possible, we report estimates of how f changes as the assumed reach r changes in Figure B4. Reassuringly, we see that the implied magnitude for f remains meaningfully interpretable as r decreases, with a maximum of $f \approx 9.45\%$ at $r = 0.1$, which would be smaller than the Fox News effect documented in DellaVigna and Kaplan (2007).

Figure B4: Blame's Persuasion Rate and Reach



Notes: The figure plots the calculated value of f for given values of $r \in [0.1, 1]$.

B.4 Virality

This appendix provides additional details on the relationship between rhetorical style and virality discussed in Section 5. We proceed in three steps. First, we describe the availability of engagement data and compare the corresponding subsample to the full tweet corpus. Second, we show that blame tweets receive more retweets and are disproportionately likely to go viral, whereas credit tweets are not. Third, we document that this pattern has strengthened over time.

Engagement data are available only for a subsample of the tweets used in the main analysis. In particular, retweet counts are observed through March 2020 and cover approximately half of all tweets in the sample over that period. Table B3 compares the full dataset with the engagement subsample. At the politician level, tweets with engagement data come from politicians who are, on average, about one year older and 2 percentage points more likely to be Republican. At the tweet level, differences are likewise modest: tweets with engagement data exhibit a slightly higher share of credit tweets, by about 4 percentage points, while the remaining dimensions are closely aligned. Overall, these comparisons suggest that the engagement subsample is broadly representative of the underlying sample, reducing concerns that the results below are driven by selection.

To study how blame and credit relate to engagement, we estimate the following specification:

$$r_{ipy t} = \alpha + \beta_1 \text{Blame}_i + \beta_2 \text{Credit}_i + \lambda_p + \mu_y + \eta_t + \varepsilon_{ipt}, \quad (6)$$

where $r_{ipy t}$ is the standardized number of retweets received by tweet i , posted by politician p in year y on topic t ; Blame_i and Credit_i are indicators equal to one if tweet i is classified as blame or credit, respectively; and λ_p , μ_y and η_t denote politician, year, and topic fixed effects. Figure B5 reports the estimates. Blame tweets receive, on average, about 0.2 standard deviations more retweets than tweets without causal rhetoric. By contrast, credit tweets do not receive significantly more retweets than non-causal tweets.

We next ask whether this relationship varies across the distribution of retweets. To do so, we re-estimate Equation 6 ten times, each time using as the dependent variable an indicator for whether the tweet falls into a given decile of the retweet distribution. Each regression therefore captures how blame and credit are associated with the probability of appearing in different parts of the retweet distribution. Figure B6 shows that blame tweets are significantly less likely to appear in the lower deciles and increasingly more likely to appear in the right tail, especially in the top decile. Credit tweets,

by contrast, are somewhat more likely to appear just above the median, but show no comparable tendency to reach the most viral part of the distribution.

Finally, we examine how the relationship between rhetorical style and virality evolves over time. We estimate:

$$y_{iptj} = \alpha + \sum_{k=2013}^{2020} \beta_1^k \left(\text{Blame}_i \times D_j^k \right) + \sum_{k=2013}^{2020} \beta_2^k \left(\text{Credit}_i \times D_j^k \right) \quad (7)$$

$$\lambda_p + \mu_t + \eta_j + \varepsilon_{iptj},$$

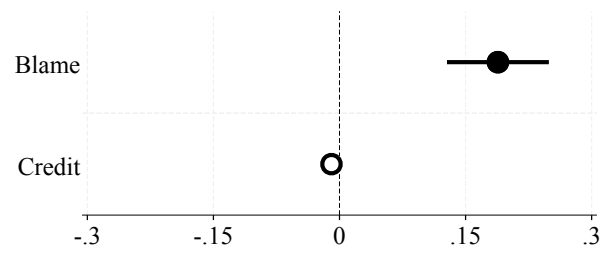
where y_{iptj} is the standardized number of retweets received by tweet i , posted by politician p on topic t in year j ; D_j^k is an indicator equal to one if the tweet was posted in year k ; and λ_p , μ_t , and η_j denote politician, topic, and year fixed effects. This specification compares the relative virality of blame and credit tweets over time, taking 2012 as the omitted year. Figure B7 presents the results. Before 2016, neither blame nor credit tweets received systematically more engagement than non-causal tweets. Starting in 2017, however, blame tweets became significantly more viral, reaching an effect of almost 0.3 standard deviations by 2020. By contrast, the relative engagement of credit tweets remained flat or declined slightly over the same period.

Table B3: Differences Between With/Without Engagement Datasets

	All Tweets until 03/2020	Tweets with Retweet Data	Difference	p-value
Female	0.275 (0.000)	0.276 (0.000)	0.001	0.196
Age	57.711 (0.007)	58.665 (0.010)	0.955	0.000
Black	0.076 (0.000)	0.074 (0.000)	-0.002	0.000
Bachelor	0.329 (0.000)	0.336 (0.000)	0.007	0.000
Master or Higher	0.626 (0.000)	0.617 (0.000)	-0.009	0.000
Republican	0.431 (0.000)	0.450 (0.000)	0.019	0.000
Nominate Score	0.434 (0.000)	0.429 (0.000)	-0.005	0.000
Share of Blame Tweets	0.144 (0.000)	0.148 (0.000)	0.004	0.000
Share of Credit Tweets	0.168 (0.000)	0.206 (0.000)	0.038	0.000
Share of None Tweets	0.687 (0.000)	0.646 (0.000)	-0.042	0.000
Share of Tweets about Economy	0.150 (0.000)	0.151 (0.000)	0.001	0.064
Share of Tweets about Environment	0.051 (0.000)	0.054 (0.000)	0.004	0.000
Share of Tweets about Gender	0.070 (0.000)	0.064 (0.000)	-0.006	0.000
Share of Tweets about Gun Control	0.032 (0.000)	0.032 (0.000)	-0.000	0.326
Share of Tweets about Healthcare	0.118 (0.000)	0.122 (0.000)	0.003	0.000
Share of Tweets about Immigration	0.055 (0.000)	0.059 (0.000)	0.003	0.000
Share of Tweets about Police	0.027 (0.000)	0.027 (0.000)	0.000	0.042
Share of Tweets about Racial Relations	0.064 (0.000)	0.059 (0.000)	-0.005	0.000
Share of Tweets about Other Topics	0.433 (0.000)	0.432 (0.000)	-0.001	0.184
Observations	2,322,957	1,178,861		

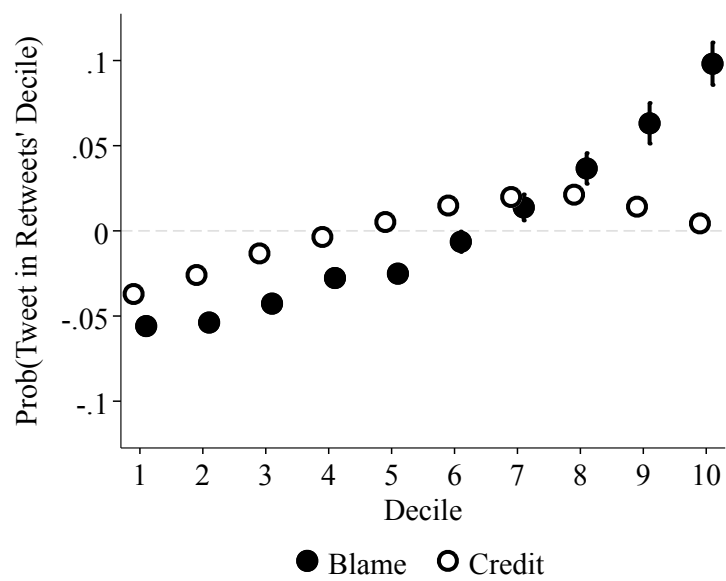
Notes: Means and standard errors in parentheses. The last column reports the p-value from a two-sided t-test of equality of means.

Figure B5: Blame, Credit, and Retweets



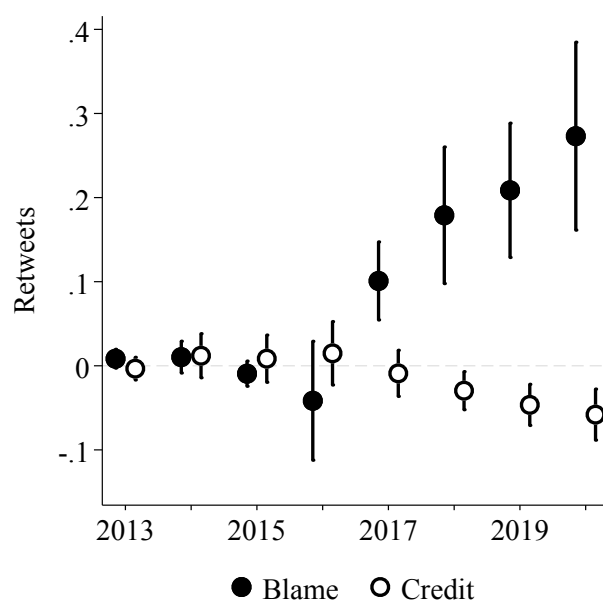
Notes: Bars represent 95 percent confidence intervals, errors clustered at the politician level.

Figure B6: Blame, Credit, and Retweets' Distribution



Notes: Bars represent 95 percent confidence intervals, errors clustered at the politician level.

Figure B7: Blame, Credit, and Engagement over Time



Notes: Shaded areas represent 95 percent confidence intervals, errors clustered at the politician level.

B.5 Assessing Economic Causal Rhetoric

This appendix provides additional details on the construction and classification of the economic-tweet sample used in Section 7. It describes the sample, the coding protocol, and the implementation of the AI-assisted classification procedure, and reports illustrative examples.

The sample used in Section 7 consists of tweets classified as blame or credit that our topic classification identifies as being about the economy. This is the same topic-classification procedure used throughout the paper and discussed in Section 4. Applying that procedure to the full tweet corpus yields 413,836 economic tweets. We do not impose any additional exclusions before classification. All tweets identified as economic are submitted to the AI-assisted coding procedure described below. A small number of requests fail because the model does not return a complete output in the required format. In practice, this concerns 379 tweets, corresponding to about 0.09 percent of the economic-tweet sample. These observations are therefore excluded from the analysis.

Each tweet is classified according to the dominant causal claim it conveys, if any. The coding task proceeds in three steps. First, for each tweet we extract the main causal claim. If no causal claim is present, the tweet is coded as *na*. Second, we identify the implied economic mechanism underlying that claim. Third, we classify the claim according to the extent to which it is supported by mainstream macroeconomic reasoning. The classifier assigns one of five raw labels: *standard*, *minor qualified*, *major qualified*, *unsupported*, and *na*. Observations coded as *na* are excluded from the analysis.²⁶

We implement the classification procedure using GPT-5 mini in batch mode under a fixed prompt and output schema. We instruct the model to rely on standard mainstream macroeconomic reasoning at the core undergraduate level because the objective of the exercise is not to adjudicate frontier debates in macroeconomics, but to assess whether a claim is supported by the settled core of economic knowledge. Using this benchmark keeps the coding transparent and reproducible, and avoids making classifications hinge on areas where expert disagreement remains substantial. The model is instructed to act as a careful research coder, to follow the codebook exactly, and to evaluate claims using standard mainstream macroeconomic reasoning. The system prompt reads as follows:

²⁶The *na* category accounts for about 14% of the economic-tweet sample. This is in line with the performance of the upstream blame–none–credit classifier used to construct the sample. In the validation set, the classifier attains an average precision of about 0.84, implying that roughly one sixth of tweets classified as blame or credit should not in fact belong to those categories. Against this benchmark, the 14% share of tweets reassigned to *na* under the stricter macroeconomic coding procedure appears broadly consistent.

You are a careful research coder.
Follow the XML codebook exactly.
Use standard mainstream bachelor-level macroeconomic reasoning.
<codebook>

The full codebook is separately reported in Listing 1. Then, the user prompt then asks the model to classify a tweet by passing the tweet identifier, text, and date in structured form.

Table B4 reports the distribution of labels in the economic-tweet sample, both pooled and separately for blame and credit tweets. Table B5 reports illustrative examples of tweets assigned to each classification category.

Table B4: Distribution of Labels

	Pooled	Split	
		Blame	Credit
Supported	9.23	8.54	9.79
Minor Qualified	61.76	46.07	74.54
Major Qualified	28.12	43.62	15.50
Unsupported	0.89	1.77	0.17

Notes: The table reports the distribution of macroeconomic-support labels in the economic-tweet sample. The pooled column reports the shares across all tweets that our topic classification identifies as being about the economy and that receive a non-*na* label under the macroeconomic coding procedure. The split columns report the same shares separately for tweets classified as blame and credit.

Table B5: Sample Tweets

Tweet	Label	Justification
I voted for additional direct payments in May and October, and tomorrow Ill vote again for \$2,000 checks to help those in need. \$600 is not enough, and Ill keep fighting for more. But thats no reason to stop a relief bill that will help people who are struggling. #SignTheBill Congress should pass my bill to get rid of the federal gas tax for the rest of the year. This is one step we can take right now to bring down high gas prices.	Standard	Cash transfers increase disposable income and consumption, easing financial hardship; standard macro effect under ordinary assumptions without major qualifiers.
Biden's \$6T budget is reckless spending that doubles down on his far-left policies. Americans can not move forward if Biden's big government lowers paychecks, raises taxes \$ harms our economy. Republicans have a way forward. URL	Minor Qualified	Recognized supply-side price effect; pass-through to consumer prices is standard but magnitude and timing depend on oil market, state taxes, retailer margins, and demand; temporary cut may have limited impact.
Extending the current spending levels is better than a government shutdown, but it is still bad policy because it damages our national security and wastes taxpayer dollars.	Major Qualified	Claim links big fiscal package to lower pay and harm; mainstream macro sees fiscal policy can affect income, taxes, labor incentives and crowding out, but effects depend on composition, financing, monetary offset, labor market frictions and timing.
	Unsupported	Tweet links extension of current spending to damage to national security and waste, but mainstream macro sees neutral/ambiguous macro effect of continuing spending; claims rely on program-specific allocation and micro/defense budget details.

Listing 1: Codebook

```
<codebook name="macro_tweet_classification">
<task>
  Classify the dominant economic causal claim in one political tweet according to how far
  mainstream bachelor-level macroeconomics would support it under the tweet's baseline
  interpretation: fully under ordinary assumptions, with only ordinary qualifications,
```

with substantial but still mainstream qualifications, not plausibly under ordinary mainstream conditions, or not at all because no meaningful economic causal claim is present.

</task>

<input>

<item name="tweet_id">Unique identifier of the tweet.</item>

<item name="tweet_text">Text of the tweet to classify.</item>

<item name="tweet_date">Date of the tweet.</item>

</input>

<claim_rules>

<rule>Code the dominant economic causal claim in the tweet.</rule>

<rule>Code causal logic, not rhetoric, morality, persuasion, or political desirability.</rule>

<rule>Interpret rhetorical language as its most reasonable economic claim, but do not invent missing detail.</rule>

<definition>A meaningful economic causal claim states or clearly implies that a policy, political, or economic factor affects an economic outcome.</definition>

<support_scope_rules>

<rule>Do not assign "na" because the claim is difficult to verify empirically, because the tweet refers to a specific policy or episode, or because the named policy is not discussed in textbooks.</rule>

<rule>If no meaningful economic causal claim is present, assign "na".</rule>

<rule>If a meaningful economic causal claim is present, assign one of "standard", "minor_qualified", "major_qualified", or "unsupported".</rule>

</support_scope_rules>

</claim_rules>

<interpretation_rules>

<definition name="baseline_interpretation">

The most reasonable ordinary reading of the tweet's dominant causal claim, without strengthening or weakening it through charitable reinterpretation.

</definition>

<definition name="ordinary_economic_assumptions">

Familiar background assumptions that normally accompany causal reasoning in mainstream undergraduate macroeconomics.

</definition>

<definition name="implied_economic_mechanism">

The abstract economic channel connecting the cause and the outcome, stated independently of the tweet's policy-specific, partisan, geographic, or branded language.

</definition>

<rule>Translate the causal claim, if any, into the implied economic mechanism before evaluating support.</rule>

<rule>Evaluate the tweet at the level of the economic mechanism, not at the level of the named policy label.</rule>

<rule>Use the specific policy name only to identify the implied economic mechanism, not as the object of support.</rule>

<rule>Do not invent a mechanism that introduces a new causal bridge not reasonably expressed in the tweet.</rule>

<rule>Do not rescue a claim merely because some logically possible economic story can be invented for it.</rule>

</interpretation_rules>

```

<knowledge_rules>
  <rule>Use standard mainstream bachelor-level macroeconomics as taught in introductory and
    intermediate undergraduate courses.</rule>
  <rule>Do not rely on niche, heterodox, or highly specialized theories unless they are part
    of standard undergraduate macroeconomic instruction.</rule>
  <rule>Evaluate the tweet's causal logic at the level of the implied economic mechanism, not
    at the level of the named policy, politician, bill, slogan, or episode.</rule>
  <rule>Do not use hindsight, ex post outcomes, or later historical developments to evaluate
    the claim.</rule>
  <rule>Do not fact-check whether the named policy or event actually produced the stated
    outcome; assess whether mainstream macroeconomics would support the causal logic.</rule>
</knowledge_rules>

<temporal_context>
  <rule>The tweet date may be used only to identify contemporaneous context and the likely
    target of the claim.</rule>
  <constraint>Do not use later events or ex post outcomes to evaluate the claim.</constraint>
</temporal_context>

<labels>
  <label name="standard">
    <definition>
      Use "standard" when mainstream bachelor-level macroeconomics would support the tweet's
        dominant causal claim under ordinary economic assumptions. The claim may still be
        simplified, but no relevant additional qualification is needed.
    </definition>
    <inclusion_criteria>
      <item>The implied mechanism is recognized in mainstream bachelor-level macroeconomics.</
        item>
      <item>The claim is supportable under ordinary, familiar assumptions in undergraduate
        macroeconomics.</item>
      <item>No relevant contextual qualification or scope restriction is needed.</item>
    </inclusion_criteria>
    <exclusion_criteria>
      <item>The claim is supportable only if important contextual conditions or scope
        restrictions do substantial work.</item>
      <item>No plausible mainstream macroeconomic condition would support the claim.</item>
    </exclusion_criteria>
  </label>

  <label name="minor_qualified">
    <definition>
      Use "minor_qualified" when the tweet's dominant causal claim maps onto a recognized
        mainstream macroeconomic mechanism and is broadly in line with its standard direction,
        but support depends on familiar contextual qualifications, scope restrictions, or
        ordinary assumptions.
    </definition>
    <inclusion_criteria>
      <item>The implied mechanism is recognized in mainstream undergraduate macroeconomics.</
        item>
      <item>The claim's direction is broadly consistent with the standard prediction.</item>
      <item>Support requires ordinary caveats about magnitude, timing, pass-through, openness,
        slack, or policy context.</item>
      <item>The qualifications refine the claim but do not substantially rewrite its baseline
        meaning.</item>
    </inclusion_criteria>
  </label>

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<exclusion_criteria>
  <item>The claim is supportable under ordinary assumptions without meaningful qualification
    .</item>
  <item>Support requires major unstated offsets or a reversal of the standard directional
    effect.</item>
  <item>No ordinary plausible mainstream condition would support the claim.</item>
</exclusion_criteria>
</label>

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<label name="major_qualified">
  <definition>
    Use "major_qualified" when the tweet's dominant causal claim can be supported by a
      recognized mainstream macroeconomic mechanism only if substantial unstated conditions,
      limiting assumptions, or offsetting forces hold, but these are still not so narrow or
      contrarian as to make the claim unsupported.
  </definition>
  <inclusion_criteria>
    <item>The implied mechanism is still recognizable within mainstream undergraduate
      macroeconomics.</item>
    <item>Support requires substantial qualifications that do heavy interpretive work.</item>
    <item>The baseline reading is not the standard prediction, but a mainstream route to
      support still exists under non-remote conditions.</item>
    <item>The claim may require specific policy composition, monetary offset, special
      incidence, or other important conditions not stated in the tweet.</item>
  </inclusion_criteria>
  <exclusion_criteria>
    <item>The claim is supportable under ordinary assumptions or only familiar caveats.</item>
    <item>Support would require narrow, unusual, strongly contrarian, or specially engineered
      assumptions.</item>
    <item>The model would need to invent a new causal bridge not reasonably expressed in the
      tweet.</item>
  </exclusion_criteria>
</label>

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<label name="unsupported">
  <definition>
    Use "unsupported" when no ordinary plausible mainstream undergraduate macroeconomic route
      supports the tweet's dominant causal claim under its baseline interpretation, or when
      support would require narrow, unusual, strongly contrarian, or specially engineered
      assumptions.
  </definition>
  <inclusion_criteria>
    <item>No ordinary plausible mainstream macroeconomic condition would support the claimed
      relationship.</item>
    <item>The claim's direction conflicts with the standard prediction and can be defended
      only by unusual or remote offsets.</item>
    <item>The implied mechanism is not recognized in mainstream undergraduate macroeconomics
      .</item>
    <item>Any defense would require rescuing the tweet beyond its ordinary meaning.</item>
  </inclusion_criteria>
</label>

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<label name="na">
  <definition>
    Use "na" when the tweet contains no meaningful economic causal claim.
  </definition>
  <inclusion_criteria>

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    <item>The tweet is descriptive, moral, partisan, rhetorical, or evaluative without a
        meaningful economic causal claim.</item>
    <item>No dominant economic causal claim can be extracted from the tweet.</item>
    <item>Any implied causal reading would require inventing details not reasonably expressed
        in the tweet.</item>
</inclusion_criteria>
<exclusion_criteria>
    <item>A recognizable economic causal claim is present.</item>
    <item>The tweet can be classified as "standard", "minor_qualified", "major_qualified", or
        "unsupported".</item>
</exclusion_criteria>
</label>
</labels>

<classification_principles>
    <rule>The labels do not measure empirical truth; they measure how far mainstream
        undergraduate macroeconomics would go in supporting the dominant causal claim under its
        baseline interpretation.</rule>
    <rule>Do not reward a claim merely because some theoretical possibility exists.</rule>
    <rule>Use "standard" when support follows under ordinary assumptions without meaningful
        qualification.</rule>
    <rule>Use "minor_qualified" when support requires familiar and ordinary qualifications that
        refine, but do not substantially alter, the tweet's baseline meaning.</rule>
    <rule>Use "major_qualified" when support requires substantial unstated qualifications or
        offsetting conditions that still remain within mainstream undergraduate macroeconomics
        and are not remote or contrarian.</rule>
    <rule>Use "unsupported" when no ordinary plausible mainstream macroeconomic condition would
        support the claim, or when support would require narrow, unusual, strongly contrarian,
        or specially engineered assumptions.</rule>
    <rule>Use "na" when no meaningful economic causal claim is present.</rule>
</classification_principles>

<language_rules>
    <rule>Write all output fields in English only.</rule>
    <rule>Do not use any non-English characters.</rule>
</language_rules>

<coding_procedure>
    <step number="1">Determine whether the tweet contains a meaningful economic causal claim. If
        not, assign "na", leave all non-applicable fields empty, and stop coding.</step>
    <step number="2">Extract the dominant causal claim in one short sentence.</step>
    <step number="3">Identify the implied economic mechanism.</step>
    <step number="4">Ask whether mainstream bachelor-level macroeconomics supports the claim
        under ordinary assumptions, supports it only with important qualifications, or does not
        plausibly support it.</step>
    <step number="5">Assign exactly one classification label: "standard", "qualified", "
        unsupported", or "na".</step>
    <step number="6">Provide a brief justification grounded in mainstream macroeconomic
        reasoning, emphasizing why support is standard, minor_qualified, major_qualified, or
        unsupported.</step>
</coding_procedure>

<output>
    <format>Return exactly one valid JSON object.</format>
    <requirements>
        <item>Return only the JSON object.</item>
        <item>Use empty strings where a field is not applicable.</item>

```

```

</requirements>
<fields>
  <field name="tweet_id">Unique identifier of the tweet.</field>
  <field name="causal_claim_extracted">The dominant causal claim in one short sentence. If
    none, return an empty string. (max 100 characters)</field>
  <field name="mechanism_identified">The implied economic mechanism in one short sentence. If
    none, return an empty string. (max 100 characters)</field>
  <field name="classification">One of: standard, minor_qualified, major_qualified,
    unsupported, na.</field>
  <field name="justification">A concise, schematic justification grounded in mainstream
    macroeconomic reasoning. Use short phrases. (max 250 characters)</field>
</fields>
</output>
</codebook>

```